



National  
Qualifications  
2026

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## **2026 Mathematics**

### **Higher - Paper 1 (Non-calculator)**

### **Question Paper Finalised Marking Instructions**

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# Marking instructions for each question

| Question                                                                                                                                                                                       |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.                                                                                                                                                                                             |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> identify common factor</li> <li>•<sup>2</sup> complete the square</li> <li>•<sup>3</sup> process for <math>r</math> and write in required form</li> </ul> <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> expand completed square form</li> <li>•<sup>2</sup> equate coefficients</li> <li>•<sup>3</sup> process for <math>q</math> and <math>r</math> and write in required form</li> </ul> | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>2(x^2 + 10x \dots</math><br/>stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>2(x+5)^2 \dots</math></li> <li>•<sup>3</sup> <math>2(x+5)^2 - 47</math></li> </ul> <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>px^2 + 2pqx + pq^2 + r</math><br/>stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>p = 2, 2pq = 20, pq^2 + r = 3</math></li> <li>•<sup>3</sup> <math>2(x+5)^2 - 47</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| 1. $2(x+5)^2 - 47$ with no working gains • <sup>1</sup> and • <sup>2</sup> only. However, see Candidate E.<br>2. In Method 2, do not penalise candidates who do not work with $p, q$ and $r$ . |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                            |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Candidate A</b><br>$2(x^2 + 10) + 3$<br>$2((x+5)^2 - 25) + 3$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$2(x+5)^2 - 47$ • <sup>3</sup> ✓<br>See exception to general marking principle (h)       |  |  | <b>Candidate B - not using required form</b><br>$px^2 + 2pqx + pq^2 + r$ • <sup>1</sup> ✓<br>$p = 2, 2pq = 20, pq^2 + r = 3$ • <sup>2</sup> ✓<br>$q = 5, r = -47$ • <sup>3</sup> ^<br><div>•<sup>3</sup> is lost as answer is not in completed square form</div>                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Candidate C</b><br>$2(x^2 + 20x) + 3$ • <sup>1</sup> ✗<br>$2((x+10)^2 - 100) + 3$ • <sup>2</sup> ✓ <sub>1</sub><br>$2(x+10)^2 - 197$ • <sup>3</sup> ✓ <sub>1</sub>                          |  |  | <b>Candidate D</b><br>$2((x+10)^2 - 100) + 3$ • <sup>1</sup> ✗ • <sup>2</sup> ✗<br>$2(x+10)^2 - 197$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Candidate E</b><br>$2(x+5)^2 - 47$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>Check: $= 2(x^2 + 10x + 25) - 47$<br>$= 2x^2 + 20x + 50 - 47$<br>$= 2x^2 + 20x + 3$ • <sup>3</sup> ✓                |  |  | <b>Candidate F</b><br>$2(x+5)^2 = 2x^2 + 20x + 50$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$\therefore 2(x+5)^2 - 47 = 2x^2 + 20x + 3$ • <sup>3</sup> ✓                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                              | Illustrative scheme                                                                                                                                                                                                                                                                               | Max mark |
|----------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> express second term in integrable form</li> <li>•<sup>2</sup> integrate first term</li> <li>•<sup>3</sup> integrate second term</li> <li>•<sup>4</sup> complete integration</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\dots + 7x^{-2}</math></li> <li>•<sup>2</sup> <math>\frac{15x^{\frac{5}{3}}}{\frac{5}{3}}</math></li> <li>•<sup>3</sup> <math>+\frac{7x^{-1}}{-1}</math></li> <li>•<sup>4</sup> <math>9x^{\frac{5}{3}} - 7x^{-1} + c</math></li> </ul> | 4        |

**Notes:**

1. •<sup>1</sup> may be implied by the appearance of  $-\frac{7x^{-1}}{1}$  or  $-\frac{7}{x}$ .
2. Do not penalise the appearance of '+c', missing integral signs or missing 'dx' at •<sup>1</sup>.
3. •<sup>3</sup> is only available for integrating a term with a negative power.
4. Do not penalise the appearance of an integral sign and/or dx throughout.
5. Do not penalise the omission of '+c' at •<sup>2</sup> or •<sup>3</sup>.
6. All coefficients must be simplified at •<sup>4</sup> stage for •<sup>4</sup> to be awarded.
7. •<sup>4</sup> is only available for attempting to integrate both terms.
8. Accept  $9x^{\frac{5}{3}} + \frac{7}{-x} + c$  for •<sup>4</sup>. However, do not accept  $9x^{\frac{5}{3}} + \frac{7}{-1x} + c$  or  $9x^{\frac{5}{3}} + -\frac{7}{x} + c$  for •<sup>4</sup>.

| Question                                                                                                                                                                                                                                                                                                | Generic scheme                                                                                                                                                                                                                                                                                                 | Illustrative scheme | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|
| 2.                                                                                                                                                                                                                                                                                                      | (continued)                                                                                                                                                                                                                                                                                                    |                     |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                |                     |          |
| <p><b>Candidate A - Integrating over two lines</b></p> $\frac{15x^{\frac{5}{3}}}{\frac{5}{3}} + 7x^{-2}$ <p>•<sup>1</sup> ✓</p> $\frac{15x^{\frac{5}{3}}}{\frac{5}{3}} + \frac{7x^{-1}}{-1}$ <p>•<sup>2</sup> ✗ •<sup>3</sup> ✓</p> $9x^{\frac{5}{3}} - 7x^{-1} + c$ <p>•<sup>4</sup> ✓<sub>1</sub></p> | <p><b>Candidate B - error in integration</b></p> $\int \left( 15x^{\frac{2}{3}} + 7x^{-2} \right) dx$ <p>•<sup>1</sup> ✓</p> $\frac{15x^{\frac{5}{3}}}{\frac{5}{3}} + \frac{7x^{-3}}{-3}$ <p>•<sup>2</sup> ✓ •<sup>3</sup> ✗</p> $9x^{\frac{5}{3}} - \frac{7}{3}x^{-3} + c$ <p>•<sup>4</sup> ✓<sub>1</sub></p> |                     |          |
| <p><b>Candidate C - +c appears on a fully simplified line of working</b></p> $9x^{\frac{5}{3}} - 7x^{-1}$ <p>•<sup>2</sup> ✓ •<sup>3</sup> ✓</p> $9\sqrt[3]{x^5} - \frac{7}{x} + c$ <p>•<sup>4</sup> ✓</p>                                                                                              | <p><b>Candidate D - simplification never fully correct</b></p> $\frac{15x^{\frac{5}{3}}}{\frac{5}{3}} + \frac{7x^{-1}}{-1}$ <p>•<sup>2</sup> ✓ •<sup>3</sup> ✓</p> $9x^{\frac{5}{3}} + \frac{7x^{-1}}{-1} + c$ $9\sqrt[5]{x^3} - \frac{7}{x} + c$ <p>•<sup>4</sup> ✗</p>                                       |                     |          |
| <p><b>Candidate E - +c not present on a fully simplified line of working</b></p> $\frac{15x^{\frac{5}{3}}}{\frac{5}{3}} + \frac{7x^{-1}}{-1} + c$ <p>•<sup>2</sup> ✓ •<sup>3</sup> ✓</p> $9x^{\frac{5}{3}} - 7x^{-1}$ <p>•<sup>4</sup> ^</p>                                                            |                                                                                                                                                                                                                                                                                                                |                     |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                  | Max mark |
|----------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> state coordinates of centre</li> <li>•<sup>2</sup> find gradient of radius</li> <li>•<sup>3</sup> state perpendicular gradient</li> <li>•<sup>4</sup> determine equation of tangent</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(-1,2)</math></li> <li>•<sup>2</sup> <math>\frac{4}{3}</math> or <math>-\frac{4}{-3}</math></li> <li>•<sup>3</sup> <math>-\frac{3}{4}</math></li> <li>•<sup>4</sup> <math>4y = -3x + 30</math></li> </ul> | 4        |

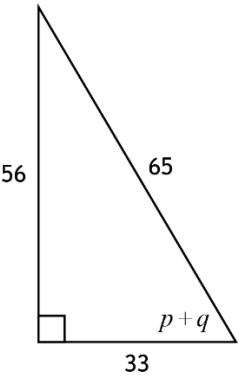
#### Notes:

1. •<sup>4</sup> is only available as a consequence of trying to find and use a perpendicular gradient.
2. At •<sup>4</sup>, accept  $3x + 4y = 30$  or any other rearrangement of the equation where the constant terms have been simplified.

#### Commonly Observed Responses:

| Candidate A - unsimplified constant terms |                  | Candidate B - $m = m_{\perp}$    |                                     |
|-------------------------------------------|------------------|----------------------------------|-------------------------------------|
| $(-1,2)$                                  | • <sup>1</sup> ✓ | $(-1,2)$                         | • <sup>1</sup> ✓                    |
| $m = \frac{-4}{-3}$                       | • <sup>2</sup> ✓ | $m = \frac{4}{3} = -\frac{3}{4}$ | • <sup>2</sup> ✓ • <sup>3</sup> ✗   |
| $m_{\perp} = -\frac{3}{4}$                | • <sup>3</sup> ✓ | $4y = -3x + 30$                  | • <sup>4</sup> ✓ <sub>1</sub> - BOD |
| $y = -\frac{3}{4}x + \frac{30}{4}$        | • <sup>4</sup> ^ | However...                       |                                     |
| However...                                |                  | $m = \frac{4}{3}$                | • <sup>2</sup> ✓                    |
| $4y = -3x + 30$                           | • <sup>4</sup> ✓ | $= -\frac{3}{4}$                 | • <sup>3</sup> ✓ - BOD              |
| $y = -\frac{3}{4}x + \frac{30}{4}$        |                  | $4y = -3x + 30$                  | • <sup>4</sup> ✓                    |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |      | Generic scheme                                                                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                                                                                                                                                                                       | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (a) |      | • <sup>1</sup> find $\sin q$                                                                                                                                                                                                     | • <sup>1</sup> $\frac{12}{13}$                                                                                                                                                                                                                                                                                                                            | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |      |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                           |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |      |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (b) | (i)  | • <sup>2</sup> find two remaining trig ratios<br><br>• <sup>3</sup> select appropriate formula and express in terms of $p$ and $q$<br><br>• <sup>4</sup> substitute into addition formula<br><br>• <sup>5</sup> find $\sin(p+q)$ | • <sup>2</sup> two from $\sin p = \frac{4}{5}$ , $\cos q = \frac{5}{13}$ ,<br>and $\cos p = \frac{3}{5}$ stated or implied<br>by • <sup>3</sup> and • <sup>4</sup><br><br>• <sup>3</sup> $\sin p \cos q + \cos p \sin q$<br><br>• <sup>4</sup> $\frac{4}{5} \times \frac{5}{13} + \frac{3}{5} \times \frac{12}{13}$<br><br>• <sup>5</sup> $\frac{56}{65}$ | 4        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     | (ii) | • <sup>6</sup> substitute into appropriate formula<br><br>• <sup>7</sup> find $\cos(p+q)$                                                                                                                                        | • <sup>6</sup> $\frac{3}{5} \times \frac{5}{13} - \frac{4}{5} \times \frac{12}{13}$<br><br>• <sup>7</sup> $-\frac{33}{65}$                                                                                                                                                                                                                                | 2        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |      |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                           |          |
| 1. Accept two from $p = \cos^{-1}\left(\frac{3}{5}\right)$ , $p = \sin^{-1}\left(\frac{4}{5}\right)$ or $q = \cos^{-1}\left(\frac{5}{13}\right)$ for • <sup>2</sup> .<br>2. Where candidates substitute an incorrect value for $\sin p$ , $\cos p$ or $\cos q$ , • <sup>4</sup> and • <sup>6</sup> may be awarded if this value has previously been stated or it can be implied by a diagram or Pythagoras calculation. See Candidate A.<br>3. Where candidates explicitly state a value for $\sin p$ , $\cos p$ or $\cos q$ , subsequent working must follow from that value for • <sup>4</sup> to be awarded.<br>4. Award • <sup>2</sup> and • <sup>3</sup> for candidates who write $\sin\left(\frac{4}{5}\right) \times \cos\left(\frac{5}{13}\right) + \cos\left(\frac{3}{5}\right) \times \sin\left(\frac{12}{13}\right)$ , • <sup>4</sup> and • <sup>5</sup> are unavailable.<br>5. For any attempt to use $\sin(p+q) = \sin p \pm \sin q$ , • <sup>4</sup> and • <sup>5</sup> are unavailable.<br>6. Do not penalise trigonometric ratios which are less than -1 or greater than 1 throughout this question.<br>7. • <sup>5</sup> is only available for an answer expressed as a single fraction. |     |      |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                           |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                         |     |      | Generic scheme                                                                                                                                                                                                                                                                                                                              | Illustrative scheme             | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------|
| 4.                                                                                                                                                                                                                                                                                                                                                                                               | (b) | (ii) | (continued)                                                                                                                                                                                                                                                                                                                                 |                                 |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                     |     |      |                                                                                                                                                                                                                                                                                                                                             |                                 |          |
| <b>Candidate A - incorrect use of Pythagoras</b><br>Missing sides of $\sqrt{7}$ and $\sqrt{162}$ . • <sup>1</sup> is unavailable.<br><br>Follow through marks are available leading to:<br>$\sin(p+q) = \frac{4\sqrt{7} + 3\sqrt{162}}{13\sqrt{7}},$ $\cos(p+q) = \frac{3\sqrt{7} - 4\sqrt{162}}{13\sqrt{7}}, \text{ and}$ $\tan(p+q) = \frac{4\sqrt{7} + 3\sqrt{162}}{3\sqrt{7} - 4\sqrt{162}}$ |     |      | <b>Candidate B - incorrect use of Pythagoras</b><br>Missing sides of 5 and $\sqrt{194}$ . • <sup>1</sup> is unavailable.<br><br>Follow through marks are available leading to:<br>$\sin(p+q) = \frac{20 + 3\sqrt{194}}{65},$ $\cos(p+q) = \frac{15 - 4\sqrt{194}}{65}, \text{ and}$ $\tan(p+q) = \frac{20 + 3\sqrt{194}}{15 - 4\sqrt{194}}$ |                                 |          |
| <b>Candidate C - not considering <math>p+q</math> is obtuse</b><br><br><br>$\cos(p+q) = \frac{33}{65}$ <div style="display: flex; align-items: center; margin-left: 200px;"> <div style="margin-right: 10px;">•<sup>6</sup> ✗</div> <div style="margin-right: 10px;">•<sup>7</sup> ✓<sub>1</sub></div> </div>  |     |      |                                                                                                                                                                                                                                                                                                                                             |                                 |          |
| 4.                                                                                                                                                                                                                                                                                                                                                                                               | (c) |      | • <sup>8</sup> find $\tan(p+q)$                                                                                                                                                                                                                                                                                                             | • <sup>8</sup> $-\frac{56}{33}$ | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                           |     |      |                                                                                                                                                                                                                                                                                                                                             |                                 |          |
| 8. Accept $-\frac{3640}{2145}$ , $-\frac{728}{429}$ or $-\frac{280}{165}$ for • <sup>8</sup> .                                                                                                                                                                                                                                                                                                   |     |      |                                                                                                                                                                                                                                                                                                                                             |                                 |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                     |     |      |                                                                                                                                                                                                                                                                                                                                             |                                 |          |
|                                                                                                                                                                                                                                                                                                                                                                                                  |     |      |                                                                                                                                                                                                                                                                                                                                             |                                 |          |

| Question                                                                      |     |      | Generic scheme                                                                                                                            | Illustrative scheme                                                                                                 | Max mark |
|-------------------------------------------------------------------------------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------|
| 5.                                                                            | (a) | (i)  | • <sup>1</sup> interpret notation<br><br>• <sup>2</sup> state expression for $f(g(x))$                                                    | • <sup>1</sup> $f(x+3)$ or $2(g(x))^2 + 5$ stated or implied by • <sup>2</sup><br><br>• <sup>2</sup> $2(x+3)^2 + 5$ | 2        |
|                                                                               |     | (ii) | • <sup>3</sup> state expression for $g(f(x))$                                                                                             | • <sup>3</sup> $(2x^2 + 5) + 3$                                                                                     | 1        |
| <b>Notes:</b>                                                                 |     |      |                                                                                                                                           |                                                                                                                     |          |
| 1. Accept $2g(x)^2 + 5$ for • <sup>1</sup> .                                  |     |      |                                                                                                                                           |                                                                                                                     |          |
| <b>Commonly Observed Responses:</b>                                           |     |      |                                                                                                                                           |                                                                                                                     |          |
| <b>Candidate A - two attempts</b><br>$f(g(x)) = 2x^2 + 5$<br>$= 2(x+3)^2 + 5$ |     |      | <b>Candidate B - <math>f(g(x))</math> and <math>g(f(x))</math> interchanged</b><br>$f(g(x)) = (2x^2 + 5) + 3$<br>$g(f(x)) = 2(x+3)^2 + 5$ |                                                                                                                     |          |
| • <sup>1</sup> ✗ • <sup>2</sup> ✗                                             |     |      | • <sup>1</sup> ✗ • <sup>2</sup> ✓ <sub>1</sub><br>• <sup>3</sup> ✓ <sub>1</sub>                                                           |                                                                                                                     |          |
|                                                                               | (b) |      | • <sup>4</sup> state range                                                                                                                | • <sup>4</sup> $g(f(x)) \geq 8$                                                                                     | 1        |
| <b>Notes:</b>                                                                 |     |      |                                                                                                                                           |                                                                                                                     |          |
| 2. Accept $y \geq 8$ or $[8, \infty)$ for • <sup>4</sup> .                    |     |      |                                                                                                                                           |                                                                                                                     |          |
| <b>Commonly Observed Responses:</b>                                           |     |      |                                                                                                                                           |                                                                                                                     |          |
|                                                                               |     |      |                                                                                                                                           |                                                                                                                     |          |



| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  | Generic scheme                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                                                                                                                | Max mark |
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| 6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> state an appropriate pathway</li> <li>•<sup>2</sup> express pathway in terms of <b>u</b>, <b>v</b>, and <b>w</b></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> eg <math>\overrightarrow{MH} + \overrightarrow{HE} + \overrightarrow{EN}</math> stated or implied by •<sup>2</sup></li> <li>•<sup>2</sup> <math>-\frac{1}{2}\mathbf{u} - \mathbf{v} + \frac{1}{3}\mathbf{w}</math></li> </ul> | 2        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                    |          |
| <p>1. Do not penalise inconsistent vector notation (for example lack of arrows or brackets).</p> <p>2. Where an invalid pathway has been stated, award 0/2.</p> <p>3. •<sup>1</sup> may be awarded for the appearance of <math>\frac{1}{2}\mathbf{u} - \mathbf{u} - \mathbf{v} + \mathbf{w} - \frac{2}{3}\mathbf{w}</math>. •<sup>2</sup> is not available for an unsimplified expression.</p> <p>4. Where no pathway has been stated, see Candidates A and B.</p> |  |  |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                    |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                    |          |
| Candidate A - no pathway stated and incorrect expression in <b>u</b> , <b>v</b> AND <b>w</b>                                                                                                                                                                                                                                                                                                                                                                       |  |  | Candidate B - no pathway stated and incorrect expression in <b>v</b> , <b>w</b> AND <b>u</b>                                                                                     |                                                                                                                                                                                                                                                                                    |          |
| $-\frac{1}{2}\mathbf{u} - \mathbf{v} \dots$                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  | $\dots - \mathbf{v} + \frac{1}{3}\mathbf{w}$                                                                                                                                     |                                                                                                                                                                                                                                                                                    |          |
| Award 1/2                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |  | Award 1/2                                                                                                                                                                        |                                                                                                                                                                                                                                                                                    |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  | Generic scheme                                                                                                                                                                                                                                     | Illustrative scheme                                                                                                                                                                                                                                                                                                         | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> write second term in differentiable form</li> <li>•<sup>2</sup> differentiate first term</li> <li>•<sup>3</sup> differentiate second term</li> <li>•<sup>4</sup> evaluate gradient</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> ... + <math>4x^{\frac{1}{2}}</math> stated or implied by •<sup>3</sup></li> <li>•<sup>2</sup> 2</li> <li>•<sup>3</sup> ... + <math>\frac{1}{2} \times 4x^{-\frac{1}{2}}</math></li> <li>•<sup>4</sup> <math>\frac{8}{3}</math> or <math>2\frac{2}{3}</math></li> </ul> | 4        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                             |          |
| <p>1. •<sup>3</sup> is only available for differentiating a term with a fractional index.</p> <p>2. •<sup>4</sup> is only available as a consequence of substituting into an expression involving a fractional power of <math>x</math>.</p> <p>3. Where candidates make no attempt to differentiate, or integrate one or other of the terms, •<sup>4</sup> is unavailable. See Candidates A, B and C.</p> <p>4. •<sup>4</sup> is not available where a candidate subsequently finds a perpendicular gradient.</p> <p>5. Accept <math>y = \frac{8}{3}</math> only where a substitution has been shown, no other values for <math>y</math> have been stated, or <math>\frac{8}{3}</math> has been used as the gradient in the equation of the tangent.</p> <p>6. Where candidates equate <math>\frac{dy}{dx}</math> to 0, •<sup>4</sup> is unavailable. See Candidate D.</p> |  |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                             |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                             |          |
| <b>Candidate A - limited evidence of differentiation of the second term</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | <b>Candidate B - second term integrated</b>                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                             |          |
| $y = 2x + 4x^{\frac{1}{2}}$<br>$\frac{dy}{dx} = 2 + 2x^{\frac{3}{2}}$<br>$2 + 2(9)^{\frac{3}{2}}$<br>$= 56$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | $y = 2x + 4x^{\frac{1}{2}}$<br>$\frac{dy}{dx} = 2 + \frac{8}{3}x^{\frac{3}{2}}$<br>$2 + \frac{8}{3}(9)^{\frac{3}{2}}$<br>$= 74$                                                                                                                    |                                                                                                                                                                                                                                                                                                                             |          |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✓ •<sup>3</sup> ✗</li> <li>•<sup>4</sup> ✓<sub>1</sub></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✓ •<sup>3</sup> ✗ •<sup>4</sup> ✗</li> </ul>                                                                                                                         |                                                                                                                                                                                                                                                                                                                             |          |
| <b>Candidate C - inclusion of + c</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | <b>Candidate D - equating to 0</b>                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                             |          |
| $y = 2x + 4x^{\frac{1}{2}}$<br>$\frac{dy}{dx} = 2 + 2x^{-\frac{1}{2}} + c$<br>$m = \frac{8}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  | $y = 2x + 4x^{\frac{1}{2}}$<br>$\frac{dy}{dx} = 2 + 2x^{-\frac{1}{2}} = 0$<br>$m = \frac{8}{3}$                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                             |          |
| <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✓ •<sup>3</sup> ✗ •<sup>4</sup> ✗</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> ✓</li> <li>•<sup>2</sup> ✓ •<sup>3</sup> ✓ •<sup>4</sup> ✗</li> </ul>                                                                                                                         |                                                                                                                                                                                                                                                                                                                             |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Max mark |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8.       |  |  | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> apply <math>\log_2 x + \log_2 y = \log_2 xy</math></li> <li>•<sup>2</sup> write in exponential form</li> <li>•<sup>3</sup> express in standard quadratic form</li> <li>•<sup>4</sup> solve quadratic and state solution of log equation</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> apply <math>\log_2 x + \log_2 y = \log_2 xy</math></li> <li>•<sup>2</sup> apply <math>m\log_2 x = \log_2 x^m</math></li> <li>•<sup>3</sup> express in standard quadratic form</li> <li>•<sup>4</sup> solve quadratic and state solution of log equation</li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_2(x(x+2)) = \dots</math></li> <li>•<sup>2</sup> <math>x(x+2) = 2^3</math></li> <li>•<sup>3</sup> <math>x^2 + 2x - 8 = 0</math></li> <li>•<sup>4</sup> <math>-4, 2</math> and <math>x &gt; 0 \Rightarrow x = 2</math></li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_2(x(x+2)) = \dots</math></li> <li>•<sup>2</sup> <math>\dots = \log_2 2^3</math></li> <li>•<sup>3</sup> <math>x^2 + 2x - 8 = 0</math></li> <li>•<sup>4</sup> <math>-4, 2</math> and <math>x &gt; 0 \Rightarrow x = 2</math></li> </ul> | 4        |

**Notes:**

1. Accept  $\log_2 x(x+2) = \dots$  or  $\log_2 x^2 + 2x = \dots$  for •<sup>1</sup>.
2. •<sup>1</sup> and •<sup>2</sup> may be awarded for  $x(x+2) = 2^3$ . However, where 'log' has been scored out, see Candidates C, D and E.
3. Each line of working must be equivalent to the line above within a valid strategy.
4. •<sup>2</sup> is not available for  $x(x+2) = 3^2$ . However, candidates may still gain •<sup>3</sup> and •<sup>4</sup>. See Candidate B.
5. •<sup>3</sup> and •<sup>4</sup> are only available if the quadratic reached at •<sup>3</sup> is obtained by applying the rules in •<sup>1</sup> and •<sup>2</sup>. See Candidates A and B.
6. •<sup>4</sup> is only available for solving a polynomial of degree two or higher.
7. At •<sup>4</sup>, accept any indication that -4 has been discarded. For example, scoring out  $x = -4$  or underlining  $x = 2$ .
8. Do not penalise the omission of  $x > 0$  at •<sup>4</sup>.

| Question                                                  |             | Generic scheme |                                                           | Illustrative scheme |  | Max mark |
|-----------------------------------------------------------|-------------|----------------|-----------------------------------------------------------|---------------------|--|----------|
| 8.                                                        | (continued) |                |                                                           |                     |  |          |
| Commonly Observed Responses:                              |             |                |                                                           |                     |  |          |
| <b>Candidate A - <math>2^3 = 6</math></b>                 |             |                | <b>Candidate B</b>                                        |                     |  |          |
| $\log_2 x(x+2) = 3$                                       |             |                | $\log_2 x(x+2) = 3$                                       |                     |  |          |
| $x(x+2) = 6$                                              |             |                | $x(x+2) = 9$                                              |                     |  |          |
| $x^2 + 2x - 6 = 0$                                        |             |                | $x^2 + 2x - 9 = 0$                                        |                     |  |          |
| $x = -1 \pm \sqrt{7}$                                     |             |                | $x = -1 \pm \sqrt{10}$                                    |                     |  |          |
| $x > 0 \Rightarrow x = \sqrt{7} - 1$                      |             |                | $x > 0 \Rightarrow x = \sqrt{10} - 1$                     |                     |  |          |
| <b>Candidate C - valid with and without scoring out</b>   |             |                | <b>Candidate D - invalid with and without scoring out</b> |                     |  |          |
| <del><math>\log_2 x(x+2) = \log_2 8</math></del>          |             |                | <del><math>\log_2 x(x+2) = \log_2 3</math></del>          |                     |  |          |
| $x(x+2) = 8$                                              |             |                | $x(x+2) = 8$                                              |                     |  |          |
| $x^2 + 2x - 8 = 0$                                        |             |                | $x^2 + 2x - 8 = 0$                                        |                     |  |          |
| $x = \cancel{4}, 2$                                       |             |                | $x = \cancel{4}, 2$                                       |                     |  |          |
| $\bullet^3$ and $\bullet^4$ are available                 |             |                | $\bullet^3$ and $\bullet^4$ are not available             |                     |  |          |
| <b>Candidate E - invalid with and without scoring out</b> |             |                | <b>Candidate F - invalid processing of logs</b>           |                     |  |          |
| <del><math>\log_2 x(x+2) = \log_2 3</math></del>          |             |                | $\log_2 x(x+2) = 2^3$                                     |                     |  |          |
| $x(x+2) = 3$                                              |             |                | $x(x+2) = 8$                                              |                     |  |          |
| $x^2 + 2x - 3 = 0$                                        |             |                | $x^2 + 2x - 8 = 0$                                        |                     |  |          |
| $x = \cancel{3}, 1$                                       |             |                | $x = \cancel{4}, 2$                                       |                     |  |          |
| $\bullet^3$ and $\bullet^4$ are not available             |             |                | $\bullet^3$ and $\bullet^4$ are not available             |                     |  |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                            | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.       | (a) |  | <p>•<sup>1</sup> find an appropriate vector eg <math>\overrightarrow{DE}</math></p> <p>•<sup>2</sup> find a second vector eg <math>\overrightarrow{EF}</math> and compare</p> <p>•<sup>3</sup> appropriate conclusion</p> | <p>•<sup>1</sup> eg <math>\overrightarrow{DE} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}</math></p> <p>•<sup>2</sup> eg <math>\overrightarrow{EF} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \therefore \overrightarrow{DE} = 2\overrightarrow{EF}</math></p> <p>•<sup>3</sup> <math>\Rightarrow</math> DE is parallel to EF (common direction)<br/>and E is a common point<br/><math>\Rightarrow</math> D, E and F are collinear</p> | 3        |

#### Notes:

1. Do not penalise inconsistent vector notation (for example lack of arrows or brackets).
2. The comparison at •<sup>2</sup> may be expressed as a ratio, for example  $DE : EF = 2 : 1$ .
3. If no comparison of vectors or the trivial comparison ' $\overrightarrow{DE} = \overrightarrow{EF}$ ' is made at •<sup>2</sup>, then •<sup>3</sup> is not available.
4. •<sup>3</sup> can only be awarded if a candidate has stated 'parallel', 'common point' and 'collinear'.
5. Candidates who state or imply that 'points are parallel' or 'vectors are collinear' or state 'parallel and share a common point  $\Rightarrow$  collinear' do not gain •<sup>3</sup>. There must be reference to the points.
6. Do not accept ' $d, e$  and  $f$  are collinear' at •<sup>3</sup>.

#### Commonly Observed Responses:

##### Candidate A - missing labels

$$\begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} \quad \bullet^1 \wedge$$

$$\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \therefore \overrightarrow{DE} = 2\overrightarrow{EF} \quad \bullet^2 \checkmark_1$$

Missing labels treated as a repeated error

$\Rightarrow$  DE is parallel to EF (common direction)  
and E is a common point  
 $\Rightarrow$  D, E and F are collinear  $\bullet^3 \checkmark_1$

##### Candidate B

$$\overrightarrow{EF} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \quad \bullet^1 \checkmark$$

$$\overrightarrow{DE} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \quad \bullet^2 \checkmark$$

$\therefore 2\overrightarrow{DE} = \overrightarrow{EF}$

ignore working subsequent to correct statement on previous line

$\Rightarrow$  DE is parallel to EF (common direction)  
and E is a common point  
 $\Rightarrow$  D, E and F are collinear  $\bullet^3 \checkmark$

| Question |     |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Max mark |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.       | (b) |  | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>4</sup> interpret ratio</li> <li>•<sup>5</sup> find coordinates of G</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>4</sup> interpret ratio</li> <li>•<sup>5</sup> find coordinates of G</li> </ul> <p><b>Method 3</b></p> <ul style="list-style-type: none"> <li>•<sup>4</sup> use section formula</li> <li>•<sup>5</sup> find coordinates of G</li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>\begin{pmatrix} 3 \\ -6 \\ 9 \end{pmatrix}</math> and <math>\begin{pmatrix} 4 \\ -8 \\ 12 \end{pmatrix}</math></li> <li>or <math>\begin{pmatrix} 7 \\ -14 \\ 21 \end{pmatrix}</math></li> <li>•<sup>5</sup> (6,-8,22)</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>4</sup> eg <math>\overrightarrow{FG} = \frac{4}{3}\overrightarrow{DF}</math> or <math>\overrightarrow{DG} = \frac{7}{3}\overrightarrow{DF}</math></li> <li>•<sup>5</sup> (6,-8,22)</li> </ul> <p><b>Method 3</b></p> <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>\frac{1}{3}(-4\mathbf{d} + 7\mathbf{f})</math> or <math>\mathbf{f} = \frac{3}{7}\mathbf{g} + \frac{4}{7}\mathbf{d}</math></li> <li>•<sup>5</sup> (6,-8,22)</li> </ul> | 2        |

**Notes:**

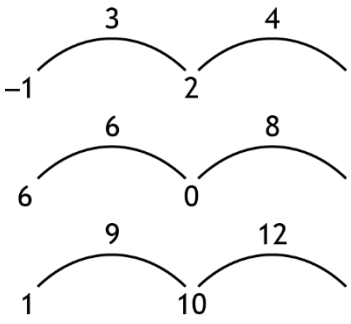
7. Do not penalise inconsistent vector notation (for example lack of arrows or brackets).

8. For (6,-8,22) without working award 2/2.

9. For  $\begin{pmatrix} 6 \\ -8 \\ 22 \end{pmatrix}$  without working award 1/2.

10. For  $\left(\frac{17}{4}, -\frac{9}{2}, \frac{67}{4}\right)$  (ratio of 4:3 with working) award 1/2. See Candidate C.

11. No marks are available for finding an internal division point, for example  $\left(\frac{2}{7}, \frac{24}{7}, \frac{34}{7}\right)$  or using other ratios.

| Question                                                                           |     |             | Generic scheme                | Illustrative scheme                                                                                                                                                          | Max mark         |
|------------------------------------------------------------------------------------|-----|-------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 9.                                                                                 | (b) | (continued) |                               |                                                                                                                                                                              |                  |
| Commonly Observed Responses:                                                       |     |             |                               |                                                                                                                                                                              |                  |
| Candidate C - ratio of 4:3                                                         |     |             |                               | Candidate D                                                                                                                                                                  |                  |
| $\overrightarrow{FG} = \frac{3}{4} \overrightarrow{DF}$                            |     |             | • <sup>4</sup> ✗              | $\frac{\overrightarrow{FG}}{\overrightarrow{DF}} = \frac{4}{3}$                                                                                                              | • <sup>4</sup> ✓ |
| $G = \left( \frac{17}{4}, -\frac{9}{2}, \frac{67}{4} \right)$                      |     |             | • <sup>5</sup> ✓ <sub>1</sub> | $3\overrightarrow{FG} = 4\overrightarrow{DF}$<br>$3(\mathbf{g} - \mathbf{f}) = 4(\mathbf{f} - \mathbf{d})$<br>$3\mathbf{g} = 7\mathbf{f} - 4\mathbf{d}$<br>$G = (6, -8, 22)$ | • <sup>5</sup> ✓ |
| Candidate E - stepping out using absolute values                                   |     |             |                               | Candidate F - Component form of section formula                                                                                                                              |                  |
|  |     |             | • <sup>4</sup> ✓              | $\frac{4(-1) + 3x}{7} = 2$<br>$\frac{4(6) + 3y}{7} = 0$<br>$\frac{4(1) + 3z}{7} = 10$                                                                                        | • <sup>4</sup> ✓ |
| $G = (6, -8, 22)$                                                                  |     |             | • <sup>5</sup> ✓              | $G = (6, -8, 22)$                                                                                                                                                            | • <sup>5</sup> ✓ |

| Question |  |  | Generic scheme                                                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                             | Max mark |
|----------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> start to integrate</li> <li>•<sup>2</sup> complete integration</li> <li>•<sup>3</sup> substitute limits</li> <li>•<sup>4</sup> evaluate integral</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>-6 \cos\left(3x - \frac{\pi}{2}\right)</math></li> <li>•<sup>2</sup> <math>\dots \times \frac{1}{3}</math></li> <li>•<sup>3</sup> <math>\left(-2 \cos\left(3 \times \frac{\pi}{3} - \frac{\pi}{2}\right)\right) - \left(-2 \cos\left(3 \times \frac{\pi}{6} - \frac{\pi}{2}\right)\right)</math></li> <li>•<sup>4</sup> 2</li> </ul> | 4        |

#### Notes:

1. Award •<sup>1</sup> for the appearance of  $-6 \cos\left(3x - \frac{\pi}{2}\right)$  regardless of any **constant** multiplier.
2. For candidates who differentiate throughout, make no attempt to integrate, or use another invalid approach – for example  $\cos\left(3x - \frac{\pi}{2}\right)^2$  or  $6 \int \sin 3x - \sin \frac{\pi}{2} dx$  – award 0/4.
3. For candidates who write  $6 - \cos(\dots)$ , no further marks are available unless the lack of brackets is corrected on the next line of working.
4. Do not penalise the inclusion of '+c' at •<sup>2</sup> or •<sup>3</sup>.
5. Do not penalise the continued appearance of the integral sign or 'dx' after integrating.
6. Accept  $\left(-2 \cos\left(\frac{\pi}{2}\right)\right) - \left(-2 \cos(0)\right)$  or  $\left(-2 \cos(3 \times 60 - 90)\right) - \left(-2 \cos(3 \times 30 - 90)\right)$  for •<sup>3</sup>.
7. •<sup>4</sup> is only available where candidates have considered both limits within a trigonometric function.
8. Where candidates use a mixture of degrees and radians, for example,  $\left(-2 \cos\left(3 \times 60^\circ - \frac{\pi}{2}\right)\right) - \left(-2 \cos\left(3 \times 30^\circ - \frac{\pi}{2}\right)\right)$ , •<sup>3</sup> is not awarded. However, •<sup>4</sup> is still available.

#### Commonly Observed Responses:

|                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Candidate A - incomplete integration</b></p> <p><math>-6 \cos\left(3x - \frac{\pi}{2}\right)</math>      •<sup>1</sup> ✓ •<sup>2</sup> ^</p> <p><math>\left(-6 \cos\left(3 \times \frac{\pi}{3} - \frac{\pi}{2}\right)\right) - \left(-6 \cos\left(3 \times \frac{\pi}{6} - \frac{\pi}{2}\right)\right)</math>      •<sup>3</sup> ✓<sub>1</sub></p> <p>6      •<sup>4</sup> ✓<sub>1</sub></p> | <p><b>Candidate B - integrating over two lines</b></p> <p><math>\int 6 \sin\left(3x - \frac{\pi}{2}\right) dx</math></p> <p><math>= -6 \cos\left(3x - \frac{\pi}{2}\right)</math>      •<sup>1</sup> ✓</p> <p><math>= -6 \cos\left(3x - \frac{\pi}{2}\right) \times \frac{1}{3}</math>      •<sup>2</sup> ✗</p> <p>•<sup>3</sup> and •<sup>4</sup> are available</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



| Question                                                                                                                                                                                                                   |             | Generic scheme                                                                                                                                                                                                                                                                                                                                                        | Illustrative scheme | Max mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|
| 10.                                                                                                                                                                                                                        | (continued) |                                                                                                                                                                                                                                                                                                                                                                       |                     |          |
| Commonly Observed Responses (continued):                                                                                                                                                                                   |             |                                                                                                                                                                                                                                                                                                                                                                       |                     |          |
| Candidate C - integrated in part                                                                                                                                                                                           |             | Candidate D - integrated in part                                                                                                                                                                                                                                                                                                                                      |                     |          |
| $6 \cos\left(3x - \frac{\pi}{2}\right) \times \frac{1}{3}$<br>• <sup>1</sup> ✗ • <sup>2</sup> ✓ <sub>1</sub><br>• <sup>3</sup> and • <sup>4</sup> are available                                                            |             | $-6 \cos\left(3x - \frac{\pi}{2}\right) \times 3$<br>• <sup>1</sup> ✓ • <sup>2</sup> ✗<br>• <sup>3</sup> and • <sup>4</sup> are available                                                                                                                                                                                                                             |                     |          |
| Candidate E - insufficient evidence of integration                                                                                                                                                                         |             | Candidate F - invalid approach                                                                                                                                                                                                                                                                                                                                        |                     |          |
| $6 \cos\left(3x - \frac{\pi}{2}\right)$<br>• <sup>1</sup> ✗ • <sup>2</sup> ^<br>• <sup>3</sup> and • <sup>4</sup> are not available                                                                                        |             | $\pm 6 \sin\left(3x - \frac{\pi}{2}\right) \times \frac{1}{3}$<br>• <sup>1</sup> ✗ • <sup>2</sup> ✗<br>• <sup>3</sup> and • <sup>4</sup> are not available                                                                                                                                                                                                            |                     |          |
| Candidate G - working in degrees before integrating                                                                                                                                                                        |             | Candidate H - BEWARE                                                                                                                                                                                                                                                                                                                                                  |                     |          |
| $\int_{30}^{60} 6 \sin(3x - 90) dx$<br>$-6 \cos(3x - 90) \times \frac{1}{3}$<br>$(-2 \cos(90)) - (-2 \cos(0))$<br>2<br>• <sup>1</sup> ✗ • <sup>2</sup> ✗<br>• <sup>3</sup> ✓ <sub>1</sub><br>• <sup>4</sup> ✓ <sub>1</sub> |             | $-6 \cos\left(3x - \frac{\pi}{2}\right) \times \frac{1}{3}$<br>$\left(-2 \cos\left(3 \times \frac{\pi}{3} - \frac{\pi}{2}\right)\right) - \left(-2 \cos\left(3 \times \frac{\pi}{6} - \frac{\pi}{2}\right)\right)$<br>$\left(-2 \cos\left(\frac{\pi}{2}\right)\right) - (-2 \cos )$<br>2<br>• <sup>1</sup> ✓ • <sup>2</sup> ✓<br>• <sup>3</sup> ✓<br>• <sup>4</sup> ✗ |                     |          |

| Question |     |     | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                               | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Max mark |
|----------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.      | (a) | (i) | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> know to use <math>x = -2</math> in synthetic division</li> <li>•<sup>2</sup> complete division, interpret result and state conclusion</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> know to substitute <math>x = -2</math></li> <li>•<sup>2</sup> complete evaluation, interpret result and state conclusion</li> </ul> | <p><b>Method 1</b></p> <p>•<sup>1</sup></p> $\begin{array}{r rrrr} -2 & 1 & 7 & 18 & 16 \\ & & 1 & & \\ \hline & 1 & 8 & 19 & 32 \end{array}$ <p>•<sup>2</sup></p> $\begin{array}{r rrrr} -2 & 1 & 7 & 18 & 16 \\ & & -2 & -10 & -16 \\ \hline & 1 & 5 & 8 & 0 \end{array}$ <p>Remainder = 0 <math>\therefore (x+2)</math> is a factor</p> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(-2)^3 + 7(-2)^2 + 18(-2) + 16</math></li> <li>•<sup>2</sup> <math>= 0 \therefore (x+2)</math> is a factor</li> </ul> | 2        |

#### Notes:

1. Communication at •<sup>2</sup> must be consistent with working at that stage i.e. candidate's working must arrive legitimately at 0 before •<sup>2</sup> can be awarded.
2. The conclusion for (a)(i) must be stated in (a)(i).
3. Accept any of the following for •<sup>2</sup>:
  - '  $f(-2) = 0$  so  $(x+2)$  is a factor '
  - 'since remainder = 0, it is a factor'
  - the '0' from any method linked to the word 'factor' by 'so', 'hence',  $\therefore$ ,  $\rightarrow$ ,  $\Rightarrow$  etc
4. Do not accept any of the following for •<sup>2</sup>:
  - double underlining the '0' or boxing the '0' without comment
  - ' $x = -2$  is a factor', '... is a root'
  - the word 'factor' only, with no link

#### Commonly Observed Responses:

##### Candidate A - grid method

|     |        |        |   |                  |
|-----|--------|--------|---|------------------|
|     | $x^2$  | $5x$   | 8 |                  |
| $x$ | $x^3$  | $5x^2$ |   | • <sup>1</sup> ✓ |
| 2   | $2x^2$ |        |   |                  |

|     |        |        |      |  |
|-----|--------|--------|------|--|
|     | $x^2$  | $5x$   | 8    |  |
| $x$ | $x^3$  | $5x^2$ | $8x$ |  |
| 2   | $2x^2$ | $10x$  | 16   |  |

with no remainder

$\therefore (x+2)$  is a factor •<sup>2</sup> ✓

##### Candidate B - grid method

|     |        |        |   |                  |
|-----|--------|--------|---|------------------|
|     | $x^2$  | $5x$   | 8 |                  |
| $x$ | $x^3$  | $5x^2$ |   | • <sup>1</sup> ✓ |
| 2   | $2x^2$ |        |   |                  |

|     |        |        |      |  |
|-----|--------|--------|------|--|
|     | $x^2$  | $5x$   | 8    |  |
| $x$ | $x^3$  | $5x^2$ | $8x$ |  |
| 2   | $2x^2$ | $10x$  | 16   |  |

$$\therefore (x+2)(x^2+5x+8) = x^3+7x+18x+16$$

$\therefore (x+2)$  is a factor •<sup>2</sup> ✓

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |      | Generic scheme                                                                                                                                                                                                                       | Illustrative scheme                                                                                                                                                                                                                                            | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (a) | (ii) | <ul style="list-style-type: none"> <li>•<sup>3</sup> use discriminant</li> <li>•<sup>4</sup> evaluate discriminant and give valid reason</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>•<sup>3</sup> <math>5^2 - 4(1)(8)</math></li> <li>•<sup>4</sup> <math>-7 &lt; 0</math>,<br/> <math>\Rightarrow x^2 + 5x + 8</math> does not factorise<br/> (so <math>(x+2)</math> is the only linear factor)</li> </ul> | 2        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |      |                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                |          |
| 5. Evidence for • <sup>3</sup> may appear in the quadratic formula.<br>6. Accept $b^2 - 4ac = -7$ for • <sup>3</sup> .<br>7. Do not penalise candidates who replace " $x^2 + 5x + 8$ " with "the quadratic factor".<br>8. At • <sup>4</sup> accept interpretations such as "no real roots", "no (further) factors" and "cannot factorise (further)" with comparison with 0.<br>9. Where the quadratic factor used in • <sup>3</sup> can be factorised, • <sup>4</sup> is not available. |     |      |                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |      |                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                |          |
| <b>Candidate C - no reference to discriminant</b><br>$-7 < 0$ • <sup>3</sup> ^<br>$\Rightarrow$ the quadratic factor does not factorise • <sup>4</sup> ^<br>$\Rightarrow (x+2)$ is the only linear factor                                                                                                                                                                                                                                                                               |     |      | <b>Candidate D - minimal reference to discriminant</b><br>$-7 < 0$ • <sup>3</sup> ^<br>discriminant $< 0$<br>$\Rightarrow$ the quadratic factor does not factorise • <sup>4</sup> ✓<br>$\Rightarrow (x+2)$ is the only linear factor |                                                                                                                                                                                                                                                                |          |
| <b>Candidate E - no evaluation of discriminant</b><br>$b^2 - 4ac < 0$ • <sup>3</sup> ^ • <sup>4</sup> ^<br>$\Rightarrow$ the quadratic factor does not factorise<br>$\Rightarrow (x+2)$ is the only linear factor                                                                                                                                                                                                                                                                       |     |      | <b>Candidate F - no evaluation of discriminant</b><br>Product = 8 and Sum = 5<br>No integer pair have this product and sum so the quadratic factor does not factorise • <sup>3</sup> ^ • <sup>4</sup> ^                              |                                                                                                                                                                                                                                                                |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                                             | Illustrative scheme                                                                                                                                                                                                                                                                                                             | Max mark |
|----------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.      | (b) |  | <ul style="list-style-type: none"> <li>•<sup>5</sup> equate curves</li> <li>•<sup>6</sup> rearrange and link to (a)</li> <li>•<sup>7</sup> find <math>x</math>-coordinate</li> <li>•<sup>8</sup> find <math>y</math>-coordinate</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>5</sup> <math>2x^3 + 20x^2 + 27x + 9 = 6x^2 - 9x - 23</math></li> <li>•<sup>6</sup> <math>2x^3 + 14x^2 + 36x + 32 = 0</math> leading to <math>2(x^3 + 7x^2 + 18x + 16) = 0</math></li> <li>•<sup>7</sup> <math>x = -2</math></li> <li>•<sup>8</sup> <math>y = 19</math></li> </ul> | 4        |

#### Notes:

10. At •<sup>6</sup> do not penalise candidates who have divided the cubic equation by 2.  
11. ‘= 0’ must appear in at least one of the equations at the •<sup>6</sup> stage.  
12. •<sup>7</sup> may not be awarded for working in part (a).  
13. Where an error has occurred at the •<sup>5</sup> or •<sup>6</sup> stage, •<sup>7</sup> and •<sup>8</sup> are still available for finding **all** the solutions (the discriminant need not be used for •<sup>8</sup> to be awarded on follow-through).  
14. Where candidates “factorise” an irreducible quadratic or use another invalid strategy, •<sup>7</sup> and •<sup>8</sup> are not available.

#### Commonly Observed Responses:

##### Candidate G - not explicitly using part (a)

$$2x^3 + 20x^2 + 27x + 9 = 6x^2 - 9x - 23 \quad \bullet^5 \checkmark$$

$$2x^3 + 14x^2 + 36x + 32 = 0$$

$$\begin{array}{r|rrrr} -2 & 2 & 14 & 36 & 32 \\ & & -4 & -20 & -32 \\ \hline & 2 & 10 & 16 & 0 \end{array}$$

$$(x+2)(2x^2 + 10x + 16) = 0 \quad \bullet^6 \checkmark$$

$$2(x+2)(x^2 + 5x + 8) = 0$$

To gain •<sup>7</sup>,  $x^2 + 5x + 8$  must also be stated in (a) or the discriminant used.

Also accept  $x = -2$  or  $x + 2 = 0$ ,

AND  $x^2 + 5x + 8 = 0$  for •<sup>6</sup>

(or without = 0 if previously included)

$$x = -2 \quad \bullet^7 \checkmark$$

$$y = 19 \quad \bullet^8 \checkmark$$

##### Candidate H - not using part (a)

$$2x^3 + 20x^2 + 27x + 9 = 6x^2 - 9x - 23 \quad \bullet^5 \checkmark$$

$$2x^3 + 14x^2 + 36x + 32 = 0$$

$$\begin{array}{r|rrrr} -2 & 2 & 14 & 36 & 32 \\ & & -4 & -20 & -32 \\ \hline & 2 & 10 & 16 & 0 \end{array}$$

$$(x+2)(2x^2 + 10x + 16) \quad \bullet^6 \checkmark$$

Also accept  $x = -2$  or  $x + 2 = 0$ ,

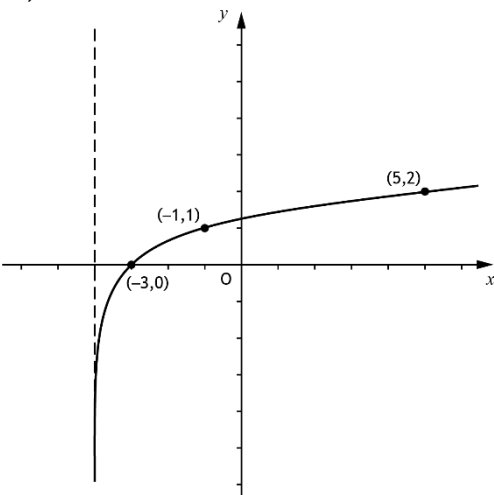
AND  $2x^2 + 10x + 16 = 0$  for •<sup>6</sup>  
(or without = 0 if previously included)

$$x = -2 \text{ and } 10^2 - 4(2)(16) = -28 < 0 \text{ so}$$

$$x = -2 \text{ is the only root} \quad \bullet^7 \checkmark$$

$$y = 19 \quad \bullet^8 \checkmark$$

| Question                                      |     |             | Generic scheme                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                                   | Max mark |
|-----------------------------------------------|-----|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.                                           | (b) | (continued) |                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                       |          |
| Commonly Observed Responses (continued):      |     |             |                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                       |          |
| Candidate I - no evidence of link to part (a) |     |             | $2x^3 + 20x^2 + 27x + 9 = 6x^2 - 9x - 23$ • <sup>5</sup> ✓<br>$2x^3 + 14x^2 + 36x + 32 = 0$ • <sup>6</sup> ^ • <sup>7</sup> ^<br>$x = -2$<br>$y = 19$ • <sup>8</sup> ✓ <sub>1</sub> | Candidate J - subtracting equations<br>$0 = (2x^3 + 20x^2 + 27x + 9) - (6x^2 - 9x - 23)$ • <sup>5</sup> ✓<br>$0 = 2x^3 + 14x^2 + 36x + 32$<br>$0 = 2(x^3 + 7x^2 + 18x + 16)$ • <sup>6</sup> ✓<br><br>However...<br>$y = (2x^3 + 20x^2 + 27x + 9) - (6x^2 - 9x - 23)$ • <sup>5</sup> ✗<br>$0 = 2(x^3 + 7x^2 + 18x + 16)$ • <sup>6</sup> ✓ <sub>1</sub> |          |

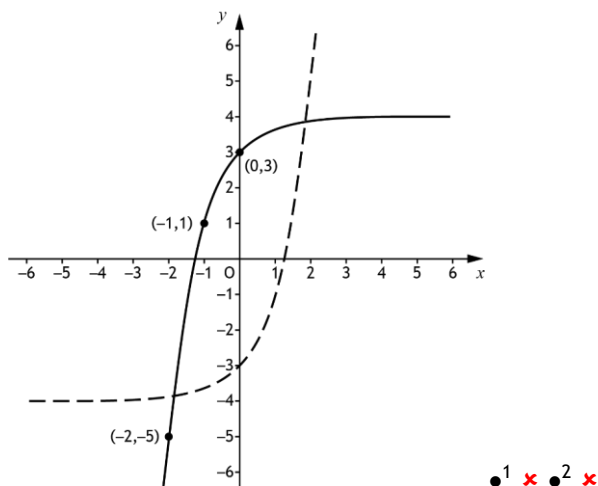
| Question |     |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                        | Illustrative scheme                                                                                                   | Max mark |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------|
| 12.      | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> graph reflected in <math>y = x</math> i.e. generally increasing for <math>-3 \leq x \leq 5</math> and passing through <math>(-3,0)</math>, <math>(-1,1)</math> and <math>(5,2)</math></li> <li>•<sup>2</sup> increasing for <math>x &lt; -3</math> (and for <math>x &gt; 5</math>) and approaching <math>x = -4</math> from the right</li> </ul> | • <sup>1</sup> , • <sup>2</sup><br> | 2        |

#### Notes:

- <sup>1</sup> is not available where the graph has a maximum turning point where  $-1 < x < 5$  and •<sup>2</sup> is not available where the graph has a maximum turning point where  $5 \leq x \leq 6$ .
- <sup>2</sup> is only available where •<sup>1</sup> has been awarded.
- The line  $x = -4$  does not need to be shown.
- For a rotation, award 0/2 - for example see Candidate A.
- Do not penalise graphs which do not extend beyond  $(5,2)$ .

#### Commonly Observed Responses:

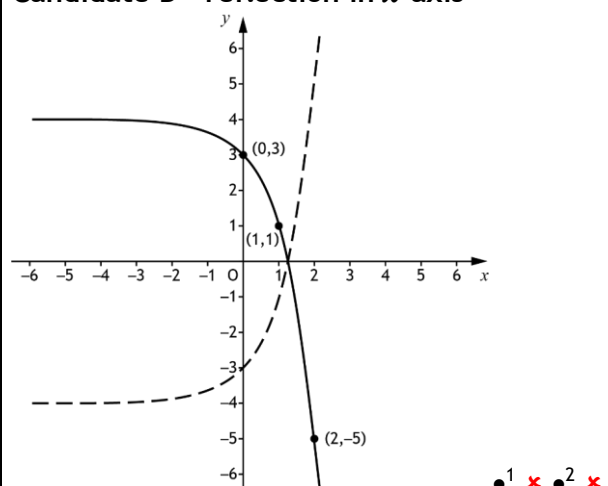
##### Candidate A - rotation about O



No further marks in this question can be inferred from this graph.

In part (b), candidates may work with the original graph and gain full marks for  $b = 4$  and  $a = 3$  (with or without working) and  $x > -4$ .

##### Candidate B - reflection in $x$ -axis



No further marks in this question can be inferred from this graph.

In part (b), candidates may work with the original graph and gain full marks for  $b = 4$  and  $a = 3$  (with or without working) and  $x > -4$ .

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |      | Generic scheme                                                             | Illustrative scheme                        | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|----------------------------------------------------------------------------|--------------------------------------------|----------|
| 12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (b) | (i)  | $\bullet^3$ determine value of $b$<br>$\bullet^4$ determine value of $a$   | $\bullet^3$ $b = 4$<br>$\bullet^4$ $a = 3$ | 2        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     | (ii) | $\bullet^5$ state domain                                                   | $\bullet^5$ $x > -4$ , ( $x \in \square$ ) | 1        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |                                                                            |                                            |          |
| 6. $a$ and $b$ may be calculated independently of each other.<br>7. The values of $a$ or $b$ can be calculated using $a = b - 1$ where $b > 1$ .<br>8. Accept ' $f(x) > -4$ ' for $\bullet^5$ .<br>9. For $\bullet^5$ , the domain can be calculated using $x > -b$ .<br>10. For $\bullet^5$ , do not penalise the omission of $x \in \square$ .<br>11. For $\bullet^5$ , the domain can be from the original graph, the candidate's transformed graph or from the value of $b$ in (b)(i). However, refer back to Candidates A and B. |     |      |                                                                            |                                            |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |      |                                                                            |                                            |          |
| <b>Candidate C - Values of <math>a</math> and <math>b</math> in equation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      | <b>Candidate D - Incorrect values of <math>a</math> and <math>b</math></b> |                                            |          |
| $\log_3(x+4)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      | $\log_3(x+4)$                                                              |                                            |          |
| $\bullet^3 \checkmark \bullet^4 \checkmark$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |      | $\bullet^3 \times$                                                         |                                            |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |      | $b = 3$                                                                    |                                            |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |      | $\bullet^4 \times$                                                         |                                            |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |      | $x > -3$                                                                   |                                            |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |      | $\bullet^5 \checkmark_1$                                                   |                                            |          |

[END OF MARKING INSTRUCTIONS]



National  
Qualifications  
2026

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## **2026 Mathematics**

### **Higher - Paper 2**

## **Question Paper Finalised Marking Instructions**

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## Marking instructions for each question

| Question                                                                                                                                                                                                                                                                               |  |  | Generic scheme                                                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                        | Max mark                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1.                                                                                                                                                                                                                                                                                     |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> use discriminant</li><li>•<sup>2</sup> apply condition and express in standard quadratic form</li><li>•<sup>3</sup> state values of <math>k</math></li></ul> | <ul style="list-style-type: none"><li>•<sup>1</sup> <math>k^2 - 4(1)(k + 3)</math></li><li>•<sup>2</sup> <math>k^2 - 4k - 12 = 0</math></li><li>•<sup>3</sup> <math>-2, 6</math></li></ul> | 3                                                                                                                           |
| Notes:                                                                                                                                                                                                                                                                                 |  |  |                                                                                                                                                                                                                  |                                                                                                                                                                                            |                                                                                                                             |
| <p>1. Where candidates state an incorrect or no condition, •<sup>2</sup> is not available. However, •<sup>3</sup> is available for finding the roots of the quadratic - see Candidate B.</p> <p>2. Where <math>x</math> appears in any expression, no further marks are available.</p> |  |  |                                                                                                                                                                                                                  |                                                                                                                                                                                            |                                                                                                                             |
| Commonly Observed Responses:                                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                  |                                                                                                                                                                                            |                                                                                                                             |
| <b>Candidate A</b><br>(For equal roots) $b^2 - 4ac = 0$<br>$k^2 - 4(1)(k + 3)$<br>$k^2 - 4k - 12$<br>$k = -2, 6$                                                                                                                                                                       |  |  | <ul style="list-style-type: none"><li>•<sup>1</sup> ✓</li><li>•<sup>2</sup> ✓</li><li>•<sup>3</sup> ✓</li></ul>                                                                                                  | <b>Candidate B</b><br>(For equal roots) $b^2 - 4ac > 0$<br>$k^2 - 4(1)(k + 3)$<br>$k^2 - 4k - 12$<br>$k = -2, 6$<br>⋮                                                                      | <ul style="list-style-type: none"><li>•<sup>2</sup> ✗</li><li>•<sup>1</sup> ✓</li><li>•<sup>3</sup> ✓<sub>1</sub></li></ul> |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |  | Generic scheme                                                                                                                                                                              | Illustrative scheme                                                                                                                                                                                                | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (a) |  | • <sup>1</sup> evaluate $\mathbf{u} \cdot \mathbf{v}$                                                                                                                                       | • <sup>1</sup> 5                                                                                                                                                                                                   | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                    |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                    |          |
| Candidate A - poor notation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                    |          |
| $\begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ -2 \\ 4 \end{pmatrix} = \begin{pmatrix} -15 \\ -4 \\ 24 \end{pmatrix} = 5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |  | • <sup>1</sup> ✗                                                                                                                                                                            |                                                                                                                                                                                                                    |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (b) |  | • <sup>2</sup> evaluate $ \mathbf{u} $<br><br>• <sup>3</sup> evaluate $ \mathbf{v} $<br><br>• <sup>4</sup> substitute into formula for scalar product<br><br>• <sup>5</sup> calculate angle | • <sup>2</sup> $ \mathbf{u}  = 7$ or $\sqrt{49}$<br><br>• <sup>3</sup> $ \mathbf{v}  = \sqrt{45}$<br><br>• <sup>4</sup> $\cos \theta = \frac{5}{7\sqrt{45}}$<br><br>• <sup>5</sup> $83.887^\circ$ OR 1.464 radians | 4        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                    |          |
| <p>1. Do not penalise candidates who treat negative signs with a lack of rigour when calculating a magnitude. For example, accept <math>\sqrt{5^2 + 2^2 + 4^2} = \sqrt{45}</math> or <math>\sqrt{-5^2 + -2^2 + 4^2} = \sqrt{45}</math> for •<sup>3</sup>. However, do not accept <math>\sqrt{5^2 - 2^2 + 4^2} = \sqrt{45}</math> for •<sup>3</sup>.</p> <p>2. •<sup>4</sup> is not available to candidates who simply state the formula <math>\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{ \mathbf{u}  \mathbf{v} }</math>. However,</p> $7 \times \sqrt{45} \times \cos \theta = 5 \text{ or } \theta = \cos^{-1}\left(\frac{5}{7\sqrt{45}}\right) \text{ is acceptable for } \bullet^4.$ <p>3. Accept correct answers rounded to <math>84^\circ</math> or 1.5 radians (or 93 gradians).</p> <p>4. Do not penalise the omission or incorrect use of units.</p> <p>5. •<sup>5</sup> is only available as a result of using a valid strategy.</p> <p>6. •<sup>5</sup> is only available for a single <b>acute</b> angle.</p> <p>7. For a correct answer with no working award 0/4.</p> |     |  |                                                                                                                                                                                             |                                                                                                                                                                                                                    |          |

| Question                                                                                                                                                                                                                                                                                                           | Generic scheme | Illustrative scheme                                                                                                                                                                                                              | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2. (b) (continued)                                                                                                                                                                                                                                                                                                 |                |                                                                                                                                                                                                                                  |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                       |                |                                                                                                                                                                                                                                  |          |
| <b>Candidate B - minimal communication</b><br>$\cos^{-1}\left(\frac{5}{\sqrt{49}\sqrt{45}}\right)$ • <sup>2</sup> ✓ • <sup>3</sup> ✓ • <sup>4</sup> ✓<br>$= 83.9^\circ$ • <sup>5</sup> ✓                                                                                                                           |                | <b>Candidate C - minimal communication</b><br>$\cos^{-1}\left(\frac{5}{\sqrt{3^2 + 2^2 + 6^2} \times \sqrt{(-5)^2 + (-2)^2 + 4^2}}\right)$ • <sup>4</sup> ✓<br>$= 83.9^\circ$ • <sup>2</sup> ✓ • <sup>3</sup> ✓ • <sup>5</sup> ✓ |          |
| <b>Candidate D - insufficient communication</b><br>• <sup>2</sup> $ \mathbf{u}  = 7$ • <sup>2</sup> ✓<br>• <sup>3</sup> $ \mathbf{v}  = \sqrt{45}$ • <sup>3</sup> ✓<br>• <sup>4</sup> $\frac{5}{7\sqrt{45}}$ • <sup>4</sup> ^<br>• <sup>5</sup> $83.887^\circ$ OR 1.464 radians      • <sup>5</sup> ✓ <sub>1</sub> |                |                                                                                                                                                                                                                                  |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Max mark |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3.       |  |  | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> equate composite function to <math>x</math></li> <li>•<sup>2</sup> write <math>g(g^{-1}(x))</math> in terms of <math>g^{-1}(x)</math></li> <li>•<sup>3</sup> state inverse function</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> write as <math>y = g(x)</math> and start to re-arrange</li> <li>•<sup>2</sup> express <math>x</math> in terms of <math>y</math></li> <li>•<sup>3</sup> state inverse function</li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>g(g^{-1}(x)) = x</math></li> <li>•<sup>2</sup> <math>\frac{1}{5}g^{-1}(x) + 6 = x</math></li> <li>•<sup>3</sup> <math>g^{-1}(x) = 5(x - 6)</math></li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>y = g(x) \Rightarrow x = g^{-1}(y)</math><br/><math>y - 6 = \frac{1}{5}x</math></li> <li>•<sup>2</sup> eg <math>x = \frac{(y - 6)}{\frac{1}{5}}</math></li> <li>•<sup>3</sup> <math>g^{-1}(y) = 5(y - 6)</math><br/><math>\Rightarrow g^{-1}(x) = 5(x - 6)</math></li> </ul> | 3        |

**Notes:**

1. In Method 1, accept  $\frac{1}{5}g^{-1}(x) + 6 = x$  for •<sup>1</sup> and •<sup>2</sup>.
2. In Method 2 accept  $y - 6 = \frac{1}{5}x$  without reference to  $y = g(x) \Rightarrow x = g^{-1}(y)$  at •<sup>1</sup>.
3. In Method 2, accept  $g^{-1}(x) = 5(x - 6)$  without reference to  $g^{-1}(y)$  at •<sup>3</sup>.
4. In Method 2, beware of candidates with working where each line is not mathematically equivalent.
5. At •<sup>3</sup> stage, accept  $g^{-1}$  written in terms of any dummy variable eg  $g^{-1}(y) = 5(y - 6)$ .
6.  $y = 5(x - 6)$  does not gain •<sup>3</sup>.
7.  $g^{-1}(x) = 5(x - 6)$  with no working gains 3/3.
8. In method 2, where candidates make multiple algebraic errors at the •<sup>1</sup> or •<sup>2</sup> stage, •<sup>3</sup> is still available. For example, see Candidate G.
9. Marks should only be awarded for using a valid strategy to find the inverse of  $g(x)$ .

| Question                                                                                                                                                                                                                                                     | Generic scheme                                                                                                                                                                   | Illustrative scheme                                                                                                                                                                                                           | Max mark                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3. (continued)</b>                                                                                                                                                                                                                                        |                                                                                                                                                                                  |                                                                                                                                                                                                                               |                                                                                                                                                                                  |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                          |                                                                                                                                                                                  |                                                                                                                                                                                                                               |                                                                                                                                                                                  |
| <b>Candidate A</b><br>$g(x) = \frac{1}{5}x + 6$<br>$y = \frac{1}{5}x + 6$ <span style="color: red;">- - -</span><br>$x = 5(y - 6)$ <span style="color: red;">- - -</span><br>$y = 5(x - 6)$ <span style="color: red;">- - -</span><br>$g^{-1}(x) = 5(x - 6)$ | <span style="color: red;">•<sup>1</sup> ✓</span> <span style="color: red;">•<sup>2</sup> ✓</span><br><span style="color: red;">•<sup>3</sup> ✗</span>                            | <b>Candidate B</b><br>$g(x) = \frac{1}{5}x + 6$<br>$y = \frac{1}{5}x + 6$ <span style="color: red;">- - -</span><br>$x = \frac{1}{5}y + 6$ <span style="color: red;">- - -</span><br>$y = 5(x - 6)$<br>$g^{-1}(x) = 5(x - 6)$ | <span style="color: red;">•<sup>1</sup> ✗</span><br><span style="color: red;">•<sup>2</sup> ✓<sub>1</sub></span><br><span style="color: red;">•<sup>3</sup> ✓<sub>1</sub></span> |
| <b>Candidate C</b><br>$g(x) = \frac{1}{5}x + 6$ <span style="color: red;">- - -</span><br>$x = \frac{1}{5}g(x) + 6$ <span style="color: red;">- - -</span><br>$g(x) = 5(x - 6)$<br>$g^{-1}(x) = 5(x - 6)$<br>The first line appears in the question          | <span style="color: red;">•<sup>1</sup> ✗</span><br><span style="color: red;">•<sup>2</sup> ✓<sub>1</sub></span><br><span style="color: red;">•<sup>3</sup> ✓<sub>1</sub></span> | <b>Candidate D - Method 1</b><br>$g(g^{-1}(x)) = \frac{1}{5}g^{-1}(x) + 6$<br>$x = \frac{1}{5}g^{-1}(x) + 6$<br>$g^{-1}(x) = 5(x - 6)$                                                                                        | <span style="color: red;">•<sup>2</sup> ✓</span><br><span style="color: red;">•<sup>1</sup> ✓</span><br><span style="color: red;">•<sup>3</sup> ✓</span>                         |
| <b>Candidate E - BEWARE</b><br>$g'(x) = \dots$<br><b>OR</b><br>$f^{-1}(x) = \dots$<br><b>OR</b><br>multiple expressions for $g^{-1}(x)$                                                                                                                      | <span style="color: red;">•<sup>3</sup> ✗</span><br><span style="color: red;">•<sup>3</sup> ✗</span><br><span style="color: red;">•<sup>3</sup> ✗</span>                         | <b>Candidate F</b><br>$x \rightarrow x \div 5 \rightarrow x \div 5 + 6 = g(x)$<br>$\div 5 \rightarrow +6$<br>$\therefore -6 \rightarrow \times 5$<br>$5(x - 6)$<br>$g^{-1}(x) = 5(x - 6)$                                     | <span style="color: red;">•<sup>1</sup> ✓</span><br><span style="color: red;">•<sup>2</sup> ✓</span><br><span style="color: red;">•<sup>3</sup> ✓</span>                         |
| <b>Candidate G - incomplete multiplication</b><br>$y = \frac{1}{5}x + 6$<br>$5y = x + 6$<br>$x = 5y - 6$<br>$g^{-1}(x) = 5x - 6$                                                                                                                             | <span style="color: red;">•<sup>1</sup> ✗</span><br><span style="color: red;">•<sup>2</sup> ✓<sub>1</sub></span><br><span style="color: red;">•<sup>3</sup> ✓<sub>1</sub></span> | <b>Candidate H - poor notation</b><br>$y = \frac{1}{5x} + 6$<br>$y - 6 = \frac{1}{5x}$<br>$x = 5(y - 6)$<br>$g^{-1}(x) = 5(x - 6)$                                                                                            | Award 2/3                                                                                                                                                                        |

| Question                                                                                                                                                                                                                                                                                                                  |  |  | Generic scheme                                                                                      | Illustrative scheme                                                                                                                                                               | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4.                                                                                                                                                                                                                                                                                                                        |  |  | <p>●<sup>1</sup> determine distance between points</p> <p>●<sup>2</sup> find equation of circle</p> | <p>●<sup>1</sup> <math>\sqrt{29}</math> stated or implied by ●<sup>2</sup></p> <p>●<sup>2</sup> <math>(x+2)^2 + (y-5)^2 = 29</math></p>                                           | 2        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                             |  |  |                                                                                                     |                                                                                                                                                                                   |          |
| <p>1. Accept <math>r^2 = 29</math> or <math>d^2 = 29</math> for ●<sup>1</sup>.</p> <p>2. ●<sup>2</sup> is only available where candidates have used a valid strategy to find the radius.</p> <p>3. ●<sup>2</sup> is not available to candidates who have a negative value for <math>r</math> or for <math>r^2</math>.</p> |  |  |                                                                                                     |                                                                                                                                                                                   |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                       |  |  |                                                                                                     |                                                                                                                                                                                   |          |
| <b>Candidate A</b><br>$(x-3)^2 + (y-7)^2 = 29$                                                                                                                                                                                                                                                                            |  |  | ● <sup>1</sup> ✓ ● <sup>2</sup> ✗                                                                   | <b>Candidate B - unsimplified answers</b><br>$(x+2)^2 + (y-5)^2 = \sqrt{29}^2$ ● <sup>1</sup> ✓ ● <sup>2</sup> ^<br>$(x-(-2))^2 + (y-5)^2 = 29$ ● <sup>1</sup> ✓ ● <sup>2</sup> ^ |          |
| <b>Candidate C - general form</b><br>$g = +2, f = -5$<br>$x^2 + y^2 + 2gx + 2fy + c = 0$<br>$3^2 + 7^2 + 2(2)(3) + 2(-5)(7) + c = 0$<br>$c = 0$<br>$x^2 + y^2 + 4x - 10y = 0$                                                                                                                                             |  |  | ● <sup>1</sup> ✓<br>● <sup>2</sup> ✓                                                                | <b>Candidate D - BEWARE</b><br>$\sqrt{3^2 - (-2)^2 + 7^2 - 5^2} = \sqrt{29}$ ● <sup>1</sup> ✗ ● <sup>2</sup> ✗<br>$\sqrt{(\pm 2)^2 + (\pm 5)^2} = \sqrt{29}$ ● <sup>1</sup> ✓     |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |  | Generic scheme                                                                                      | Illustrative scheme                                                                          | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------|
| 5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (a) |  | • <sup>1</sup> find value of $u_1$                                                                  | • <sup>1</sup> 7.25 or $\frac{29}{4}$                                                        | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |  |                                                                                                     |                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  |                                                                                                     |                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (b) |  | • <sup>2</sup> communicate condition                                                                | • <sup>2</sup> $1.125 > 1$                                                                   | 1        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |  |                                                                                                     |                                                                                              |          |
| <p>1. For •<sup>2</sup>, accept statements such as:</p> <ul style="list-style-type: none"> <li>• ‘1.125 is greater than 1’</li> <li>• ‘1.125 does not fall between -1 and 1’</li> <li>• ‘<math>-1 &lt; 1.125 &lt; 1</math> is false’</li> <li>• ‘1.125 is not in <math>-1 &lt; a &lt; 1</math>’.</li> </ul> <p>2. •<sup>2</sup> is not available for statements such as:</p> <ul style="list-style-type: none"> <li>• ‘<math>1.125 \geq 1</math>’</li> <li>• ‘It is greater than 1’</li> <li>• ‘1.125 does not fall between 0 and 1’</li> <li>• ‘<math>-1 &lt; 1.125 \not&lt; 1</math>’</li> <li>• ‘<math>-1 &lt; 1.125 &gt; 1</math>’.</li> </ul> <p>3. Candidates who state ‘<math>a &gt; 1</math>’ can only gain •<sup>2</sup> if it is explicitly stated that <math>a = 1.125</math>.</p> |     |  |                                                                                                     |                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  |                                                                                                     |                                                                                              |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (c) |  | • <sup>3</sup> find value of $u_4$<br><br>• <sup>4</sup> find value of $u_5$ and state value of $n$ | • <sup>3</sup> $u_4 = 27.275 \dots$<br><br>• <sup>4</sup> $u_5 = 35.685 \dots$ , AND $n = 5$ | 2        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |  |                                                                                                     |                                                                                              |          |
| <p>4. The value of <math>u_4</math> may appear within a substitution to find <math>u_5</math>.</p> <p>5. Do not penalise truncated or rounded values.</p> <p>6. For <math>n = 5</math> with no working, award 0/2.</p> <p>7. Where <math>n = 5</math> has not been explicitly stated, •<sup>4</sup> may be awarded where <math>u_5</math> has been explicitly compared to 30. See Candidate A.</p>                                                                                                                                                                                                                                                                                                                                                                                            |     |  |                                                                                                     |                                                                                              |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  |                                                                                                     |                                                                                              |          |
| <b>Candidate A - not stating value of <math>n</math></b><br>$u_5 = 35.685$ AND $u_5 > 30$ • <sup>4</sup> ✓<br>$u_5 = 35.685 > 30$ • <sup>4</sup> ✓<br><u><math>u_5 = 35.685</math></u> • <sup>4</sup> ^                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  |                                                                                                     |                                                                                              |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                 | Illustrative scheme      | Max mark |
|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------|
| 6.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> vertical reflection in <math>x</math>-axis identifiable from stationary points of graph</li> <li>•<sup>2</sup> vertical translation of '+1' units identifiable from stationary points of graph</li> <li>•<sup>3</sup> transformation applied in the correct order and remaining points identifiable from graph</li> </ul> | <p>•<sup>1,2,3</sup></p> | 3        |

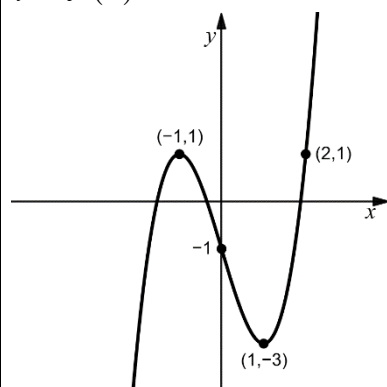
#### Notes:

1. Where candidates do not sketch a cubic function of the correct orientation, award 0/3. However, see Candidate A.
2. Where either stationary point has been incorrectly transformed, no marks are available. However, see Candidates A and B and the table for possible exceptions.
3. Do not penalise graphs which cross the  $x$ -axis at (2,0) or which do not extend past (2,1).

#### Commonly Observed Responses:

##### Candidate A - one transformation

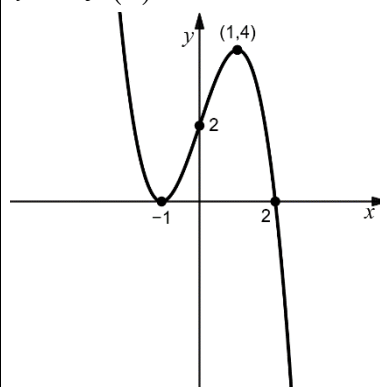
$$y = f(x) + 1$$



Award 1/3

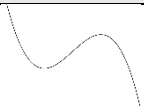
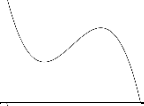
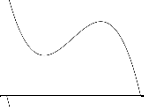
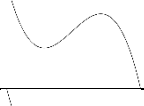
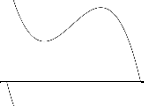
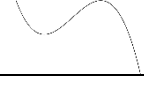
##### Candidate B - one transformation

$$y = -f(x)$$



Award 1/3



| Question                                 | Generic scheme                      | Illustrative scheme          | Max mark                                                                              |            |
|------------------------------------------|-------------------------------------|------------------------------|---------------------------------------------------------------------------------------|------------|
| 6. (continued)                           |                                     |                              |                                                                                       |            |
| Commonly Observed Responses (continued): |                                     |                              |                                                                                       |            |
| Table showing two transformations...     |                                     |                              |                                                                                       |            |
| Function                                 | Transformation of stationary points | Transformation of intercepts | Shape                                                                                 | Award      |
| $y = -f(x) - 1$                          | $(-1, -1)$ and $(1, 3)$             | $(2, -1)$ and $(0, 1)$       |    | <b>2/3</b> |
| $y = f(-x) + 1$                          | $(1, 1)$ and $(-1, -3)$             | $(-2, 1)$ and $(0, -1)$      |    | <b>1/3</b> |
| $y = -f(x + 1)$                          | $(-2, 0)$ and $(0, 4)$              | $(1, 0)$ and $(-1, 2)$       |    | <b>1/3</b> |
| $y = -f(x - 1)$                          | $(0, 0)$ and $(2, 4)$               | $(3, 0)$ and $(1, 2)$        |    | <b>1/3</b> |
| $y = f((-x) + 1)$                        | $(2, 0)$ and $(0, -4)$              | $(-1, 0)$ and $(1, -2)$      |   | <b>0/3</b> |
| $y = f((-x) - 1)$                        | $(0, 0)$ and $(-2, -4)$             | $(-3, 0)$ and $(-1, -2)$     |  | <b>0/3</b> |

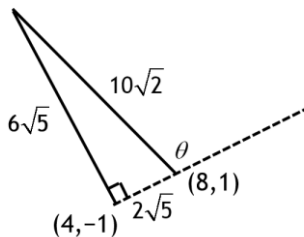
| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Max mark |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.       |  |  | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> state integral which represents the area</li> <li>•<sup>2</sup> integrate valid expression</li> <li>•<sup>3</sup> substitute limits</li> <li>•<sup>4</sup> evaluate area</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> state integral which is related to the area</li> <li>•<sup>2</sup> integrate valid expression</li> <li>•<sup>3</sup> substitute limits</li> <li>•<sup>4</sup> evaluate integral and state area</li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> eg <math>-\int_{-1}^2 (x^2 - x - 2) dx</math></li> </ul> <p>OR</p> $\int_{-1}^2 (-x^2 + x + 2) dx$ <ul style="list-style-type: none"> <li>•<sup>2</sup> eg <math>-\frac{x^3}{3} + \frac{x^2}{2} + 2x</math></li> <li>•<sup>3</sup> <math>\left( -\frac{(2)^3}{3} + \frac{(2)^2}{2} + 2(2) \right) - \left( -\frac{(-1)^3}{3} + \frac{(-1)^2}{2} + 2(-1) \right)</math></li> <li>•<sup>4</sup> <math>\frac{9}{2} (\text{units}^2)</math></li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> eg <math>\int_{-1}^2 (x^2 - x - 2) dx</math></li> <li>•<sup>2</sup> eg <math>\frac{x^3}{3} - \frac{x^2}{2} - 2x</math></li> <li>•<sup>3</sup> <math>\left( \frac{2^3}{3} - \frac{2^2}{2} - 2(2) \right) - \left( \frac{(-1)^3}{3} - \frac{(-1)^2}{2} - 2(-1) \right)</math></li> <li>•<sup>4</sup> <math>-\frac{9}{2} \therefore \text{area} = \frac{9}{2} (\text{units}^2)</math></li> </ul> | 4        |

#### Notes:

1. •<sup>1</sup> is not available to candidates who omit 'dx'.
2. Limits must appear at the •<sup>1</sup> stage for •<sup>1</sup> to be awarded.
3. In Method 1, accept  $\int_{-1}^2 (0) - (x^2 - x - 2) dx$  OR  $\int_2^{-1} (x^2 - x - 2) dx$  for •<sup>1</sup>.
4. Where candidates differentiate one or more terms at •<sup>2</sup>, then •<sup>3</sup> and •<sup>4</sup> are unavailable.
5. Candidates who substitute limits without integrating do not gain •<sup>3</sup> and •<sup>4</sup>.
6. Do not penalise the inclusion of '+c' at •<sup>2</sup> and •<sup>3</sup>.
7. Do not penalise the continued appearance of the integral sign or 'dx' after •<sup>1</sup>.
8. In Method 2, where a candidate's integral leads to a positive value, •<sup>4</sup> is not available.

| Question                                                                                                                                                                                                                                 | Generic scheme                                                                              | Illustrative scheme                                                                                                                                                      | Max mark                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>7. (continued)</b>                                                                                                                                                                                                                    |                                                                                             |                                                                                                                                                                          |                                                                                             |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                      |                                                                                             |                                                                                                                                                                          |                                                                                             |
| <b>Candidate A - missing working</b><br>$-\int_{-1}^2 (x^2 - x - 2)$ $= -\frac{x^3}{3} + \frac{x^2}{2} - 2x$ $= \frac{9}{2}$                                                                                                             | <p>•<sup>1</sup> ^</p> <p>•<sup>2</sup> ✓ •<sup>3</sup> ^</p> <p>•<sup>4</sup> ✓</p>        | <b>Candidate B - evidence of substitution using a calculator</b><br>$= -\frac{x^3}{3} + \frac{x^2}{2} - 2x$ $= \frac{10}{3} - \left(-\frac{7}{6}\right)$ $= \frac{9}{2}$ | <p>•<sup>1</sup> ^</p> <p>•<sup>2</sup> ✓</p> <p>•<sup>3</sup> ✓</p> <p>•<sup>4</sup> ✓</p> |
| <b>Candidate C - communication in Method 2</b><br>$\vdots$ $= -\frac{9}{2}$ <p>Hence area is <math>\frac{9}{2}</math></p>                                                                                                                | <p>•<sup>4</sup> ✓</p>                                                                      | <b>Candidate D - communication in Method 2</b><br>$\vdots$ $= -\frac{9}{2}$ $= \frac{9}{2} (\text{units}^2)$                                                             | <p>•<sup>4</sup> ✗</p>                                                                      |
| $= -\frac{9}{2} \text{ but area can't be negative}$ <p>so <math>\frac{9}{2} (\text{units}^2)</math></p>                                                                                                                                  | <p>•<sup>4</sup> ✓</p>                                                                      | $= -\frac{9}{2} \text{ but area can't be negative}$ $= \frac{9}{2} (\text{units}^2)$                                                                                     | <p>•<sup>4</sup> ✗</p>                                                                      |
| $= -\frac{9}{2}$ <p>so <math>\frac{9}{2} (\text{units}^2)</math> as it's under the <math>x</math>-axis</p>                                                                                                                               | <p>•<sup>4</sup> ✓</p>                                                                      | $= -\frac{9}{2}$ <p>so <math>\frac{9}{2} (\text{units}^2)</math></p>                                                                                                     | <p>•<sup>4</sup> ✗</p>                                                                      |
| <b>Candidate E - limits not stated initially</b><br>$= -\frac{x^3}{3} + \frac{x^2}{2} - 2x$ $= \left( -\frac{(2)^3}{3} + \frac{(2)^2}{2} + 2(2) \right)$ $- \left( -\frac{(-1)^3}{3} + \frac{(-1)^2}{2} + 2(-1) \right)$ $= \frac{9}{2}$ | <p>•<sup>1</sup> ^</p> <p>•<sup>2</sup> ✓</p> <p>•<sup>3</sup> ✓</p> <p>•<sup>4</sup> ✓</p> |                                                                                                                                                                          |                                                                                             |

| Question                                                                                                                                                                                                                                                                                        |     |  | Generic scheme                                                                                                                                                                                                                             | Illustrative scheme                                                                                                                                                                                                                  | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8.                                                                                                                                                                                                                                                                                              | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> find the midpoint of AB</li> <li>•<sup>2</sup> calculate <math>m_{AB}</math></li> <li>•<sup>3</sup> state perpendicular gradient</li> <li>•<sup>4</sup> find equation of L</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(4, -1)</math></li> <li>•<sup>2</sup> <math>-2</math></li> <li>•<sup>3</sup> <math>\frac{1}{2}</math></li> <li>•<sup>4</sup> <math>y = \frac{1}{2}x - 3</math></li> </ul> | 4        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                   |     |  |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                      |          |
| 1. • <sup>4</sup> is only available to candidates who find and use a perpendicular gradient and a midpoint.<br>2. The gradient of the perpendicular bisector must appear in fully simplified form at • <sup>3</sup> or • <sup>4</sup> stage for • <sup>4</sup> to be awarded - see Candidate A. |     |  |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                             |     |  |                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                      |          |
| <b>Candidate A - unsimplified gradient</b>                                                                                                                                                                                                                                                      |     |  | <b>Candidate B - using midpoint and <math>m_{AC}</math></b>                                                                                                                                                                                |                                                                                                                                                                                                                                      |          |
| $m = -\frac{24}{12}$                                                                                                                                                                                                                                                                            |     |  | $(4, -1)$                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                      |          |
|                                                                                                                                                                                                                                                                                                 |     |  | $m_{AC} = -1$                                                                                                                                                                                                                              |                                                                                                                                                                                                                                      |          |
| $m_{\perp} = \frac{12}{24}$                                                                                                                                                                                                                                                                     |     |  | $m_{\perp} = 1$                                                                                                                                                                                                                            |                                                                                                                                                                                                                                      |          |
| $24y = 12x - 72$                                                                                                                                                                                                                                                                                |     |  | $y = x - 5$                                                                                                                                                                                                                                |                                                                                                                                                                                                                                      |          |
| <b>Candidate C</b>                                                                                                                                                                                                                                                                              |     |  | <b>Candidate D</b>                                                                                                                                                                                                                         |                                                                                                                                                                                                                                      |          |
| $m = -2 = \frac{1}{2}$                                                                                                                                                                                                                                                                          |     |  | $m = -2$                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                      |          |
|                                                                                                                                                                                                                                                                                                 |     |  | $= \frac{1}{2}$                                                                                                                                                                                                                            |                                                                                                                                                                                                                                      |          |
| $y = \frac{1}{2}x - 3$                                                                                                                                                                                                                                                                          |     |  | $y = \frac{1}{2}x - 3$                                                                                                                                                                                                                     |                                                                                                                                                                                                                                      |          |

| Question                                                                                                                                                                                                                                                                                                                                              |     |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Max mark                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 8.                                                                                                                                                                                                                                                                                                                                                    | (b) |  | <p><b>Method 1</b></p> <ul style="list-style-type: none"><li>•<sup>5</sup> calculate angle between one line and the <math>x</math>-axis</li><li>•<sup>6</sup> calculate angle between second line and the <math>x</math>-axis <b>AND</b> calculate angle between L and AC</li></ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"><li>•<sup>5</sup> calculate one angle within small triangle</li><li>•<sup>6</sup> calculate second angle within small triangle <b>AND</b> calculate angle between L and AC</li></ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"><li>•<sup>5</sup> <math>26.565...^\circ</math> OR <math>135^\circ</math></li><li>•<sup>6</sup> <math>26.565...^\circ</math> OR <math>135^\circ</math> leading to <math>(135 - 26.565... =) 108.434...^\circ</math></li></ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"><li>•<sup>5</sup> <math>26.565...^\circ</math> OR <math>45^\circ</math></li><li>•<sup>6</sup> <math>26.565...^\circ</math> OR <math>45^\circ</math> leading to <math>(180 - 45 - 26.565... =) 108.434...^\circ</math></li></ul> | 2                                        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                         |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| 3. Answers may be in radians: $0.4636...$ , and $\frac{3\pi}{4}$ ( $2.356...$ ) leading to $1.8925...$ radians.                                                                                                                                                                                                                                       |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| 4. Do not penalise the omission of units at • <sup>6</sup> .                                                                                                                                                                                                                                                                                          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| 5. Accept any answers which round to $108^\circ$ or $1.9$ radians.                                                                                                                                                                                                                                                                                    |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| 6. For $108^\circ$ without working award 0/2.                                                                                                                                                                                                                                                                                                         |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                   |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| <b>Candidate E - Incorrect working</b>                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Candidate F - Incorrect working</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                          |
| $\tan^{-1}(-1) = 45^\circ$                                                                                                                                                                                                                                                                                                                            |     |  | • <sup>5</sup> ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\tan(\theta) = -1$<br>$\theta = 45^\circ$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | • <sup>5</sup> ✗                         |
| <b>Candidate G - Poor notation</b>                                                                                                                                                                                                                                                                                                                    |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Candidate H - insufficient communication</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| $m = -1$<br>$m = \tan \theta$<br>$\theta = \tan^{-1}(1)$<br>$\theta = 45$<br>$\theta = 180 - 45 = 135$                                                                                                                                                                                                                                                |     |  | • <sup>5</sup> ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\theta = \tan^{-1}(1)$<br>$\theta = 45$<br>$180 - 45 = 135$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | • <sup>5</sup> ^                         |
| <b>Candidate I - Using point of intersection, Pythagoras and right-angled trigonometry</b>                                                                                                                                                                                                                                                            |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Candidate J - Representing gradients as vectors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                          |
| <p>A(-2,11)</p>  <p>•<sup>5</sup> ✓</p> <p>•<sup>6</sup> ✓</p> <p><math>\theta = 180 - \sin^{-1}\left(\frac{6\sqrt{5}}{10\sqrt{2}}\right) = 108.4^\circ</math></p> <p>Only two of the distances are required in the diagram and other trig ratios may be used.</p> |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$<br>$\cos \theta = \frac{-1}{\sqrt{5}\sqrt{2}}$<br>$\theta = 108.4^\circ$<br><br>Any multiples of the vectors may be used                                                                                                                                                                                                                                                                                                                                               | • <sup>5</sup> ✓<br><br>• <sup>6</sup> ✓ |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                   | Generic scheme | Illustrative scheme | Max mark |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|----------|
| 8. (b) (continued)                                                                                                                                                                                                                                                                                                                                                                                                                         |                |                     |          |
| Commonly Observed Responses (continued):                                                                                                                                                                                                                                                                                                                                                                                                   |                |                     |          |
| <p>Candidate K - Using a formula for angle between lines</p> <p>Acute angle between 2 lines = <math>\tan^{-1}\left(\frac{m_1 - m_2}{1 + m_1 m_2}\right)</math></p> <p><math>= \tan^{-1}\left(\frac{\frac{1}{2} - (-1)}{1 + (-1)\left(\frac{1}{2}\right)}\right)</math> •<sup>5</sup> ✓</p> <p><math>= \tan^{-1}(3)</math></p> <p><math>= 71.6^\circ</math></p> <p>Obtuse angle = <math>180 - 71.6 = 108.4^\circ</math> •<sup>6</sup> ✓</p> |                |                     |          |

| Question |     |   | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Max mark |     |   |     |   |     |         |   |   |   |   |   |   |
|----------|-----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|---|-----|---|-----|---------|---|---|---|---|---|---|
| 9.       |     |   | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> differentiate one non-constant term</li> <li>•<sup>2</sup> complete differentiation and interpret condition</li> <li>•<sup>3</sup> determine zeros of quadratic expression</li> <li>•<sup>4</sup> state range with justification</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> differentiate one non-constant term</li> <li>•<sup>2</sup> complete differentiation and determine zeros of quadratic expression</li> <li>•<sup>3</sup> construct nature table(s)</li> <li>•<sup>4</sup> interpret sign of derivative and state range</li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>12x^2 \dots</math> OR <math>\dots -24x</math></li> <li>•<sup>2</sup> <math>12x^2 - 24x &lt; 0</math></li> <li>•<sup>3</sup> 0 AND 2</li> <li>•<sup>4</sup> <math>0 &lt; x &lt; 2</math><br/>with eg labelled sketch: </li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>12x^2 \dots</math> OR <math>\dots -24x</math></li> <li>•<sup>2</sup> <math>12x^2 - 24x</math> AND 0 and 2 .</li> <li>•<sup>3</sup> <table border="1" data-bbox="901 996 1316 1176"> <tr> <td></td><td>...</td><td>0</td><td>...</td><td>2</td><td>...</td></tr> <tr> <td><math>f'(x)</math></td><td>+</td><td>0</td><td>-</td><td>0</td><td>+</td></tr> </table> </li> <li>•<sup>4</sup> decreasing when <math>f'(x) &lt; 0</math> so <math>0 &lt; x &lt; 2</math></li> </ul> |          | ... | 0 | ... | 2 | ... | $f'(x)$ | + | 0 | - | 0 | + | 4 |
|          | ... | 0 | ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ...      |     |   |     |   |     |         |   |   |   |   |   |   |
| $f'(x)$  | +   | 0 | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | +        |     |   |     |   |     |         |   |   |   |   |   |   |

#### Notes:

- At •<sup>3</sup>, do not penalise candidates who fail to extract the common factor or who have divided the quadratic inequality by 12.
- <sup>3</sup> and •<sup>4</sup> are not available to candidates who arrive at a linear expression at •<sup>2</sup>.
- Accept the appearance of 0 and 2 within inequalities for •<sup>3</sup>.
- In Method 2, •<sup>3</sup> and •<sup>4</sup> are not available where only one stationary point has been considered.
- At •<sup>4</sup>, accept " $x > 0, x < 2$ " together with the required justification.

#### Commonly Observed Responses:

|                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Candidate A</b></p> <p><math>12x^2 - 24x &lt; 0</math>                      •<sup>1</sup> ✓ •<sup>2</sup> ✓</p> <p><math>12x^2 - 24x = 0</math>                      •<sup>3</sup> ✓</p> <p><math>x = 0, 2</math>                                  •<sup>4</sup> ✓</p> <p><math>0 &lt; x &lt; 2</math> with sketch</p> | <p><b>Candidate B - no initial inequation</b></p> <p><math>12x^2 - 24x = 0</math>                      •<sup>1</sup> ✓ •<sup>2</sup> ✗</p> <p><math>x = 0, 2</math>                                  •<sup>3</sup> ✓</p> <p><math>0 &lt; x &lt; 2</math> with sketch              •<sup>4</sup> ✗</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Question                                                                               | Generic scheme                              | Illustrative scheme                                                                                                                             | Max mark                                                                                         |
|----------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 9. (continued)                                                                         |                                             |                                                                                                                                                 |                                                                                                  |
| Commonly Observed Responses (continued):                                               |                                             |                                                                                                                                                 |                                                                                                  |
| <b>Candidate C</b><br>Decreasing when $f'(x) < 0$<br>$f'(x) = 12x^2 - 24x$<br>$\vdots$ | $\bullet^1 \checkmark \bullet^2 \checkmark$ | <b>Candidate D - condition applied after simplification</b><br>$f'(x) = 12x^2 - 24x$<br>$x^2 - 2x < 0$<br>$x = 0, 2$<br>$0 < x < 2$ with sketch | $\bullet^1 \checkmark$<br>$\bullet^2 \wedge$<br>$\bullet^3 \checkmark$<br>$\bullet^4 \checkmark$ |



| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Max mark |
|----------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use double angle formula to express equation in terms of <math>\cos x^\circ</math></li> <li>•<sup>2</sup> arrange in standard quadratic form</li> <li>•<sup>3</sup> factorise or use quadratic formula</li> <li>•<sup>4</sup> solve for <math>\cos x^\circ</math></li> <li>•<sup>5</sup> solve for <math>x</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>3(2\cos^2 x^\circ - 1) + 10\cos x^\circ (= 1)</math></li> <li>•<sup>2</sup> <math>6\cos^2 x + 10\cos x - 4 = 0</math></li> <li>•<sup>3</sup> <math>2(3\cos x^\circ - 1)(\cos x^\circ + 2)(= 0)</math></li> <li>OR</li> <li><math display="block">\cos x = \frac{-10 \pm \sqrt{196}}{12}</math></li> <li>•<sup>4</sup> <math>\cos x^\circ = \frac{1}{3}</math>      •<sup>5</sup> <math>\cos x^\circ = -2</math></li> <li>•<sup>5</sup> 70.52..., 289.47... 'no solutions'</li> </ul> | 5        |

#### Notes:

1. Do not penalise the omission of the degree sign.
2. In the event of  $\cos^2 x^\circ - \sin^2 x^\circ$  or  $1 - 2\sin^2 x^\circ$  being substituted for  $\cos 2x^\circ$ , •<sup>1</sup> cannot be awarded until the equation reduces to a quadratic in  $\cos x^\circ$ .
3. Substituting  $2\cos^2 A - 1$  or  $2\cos^2 \alpha - 1$  for  $\cos 2x^\circ$  at the •<sup>1</sup> stage should be treated as bad form only where the equation is written in terms of  $x$  at •<sup>2</sup> stage. Otherwise, •<sup>1</sup> is not available.
4. “= 0” must appear by •<sup>3</sup> stage for •<sup>2</sup> to be awarded. However, for candidates using the quadratic formula to solve the equation, “= 0” must appear at the •<sup>2</sup> stage for •<sup>2</sup> to be awarded.
5. The equation  $6\cos^2 x^\circ - 4 + 10\cos x^\circ = 0$  does not gain •<sup>2</sup> unless •<sup>3</sup> has been awarded.
6. •<sup>3</sup> can be awarded for identifying the factors of the quadratic obtained at •<sup>2</sup>.  
For example, ‘ $6\cos x^\circ - 2 = 0$  and  $\cos x^\circ + 2 = 0$ ’.
7. Do not penalise candidates who fail to extract a common factor or who have divided the equation by 2 at •<sup>3</sup>.
8. Candidates may express their equation, obtained at •<sup>2</sup>, in the form  $6c^2 + 10c - 4 = 0$  or  $6x^2 + 10x - 4 = 0$ . In these cases, award •<sup>3</sup> for  $2(3c - 1)(c + 2) = 0$ ,  $2(3x - 1)(x + 2) = 0$  or equivalent. However, •<sup>4</sup> is only available if  $\cos x^\circ$  appears explicitly at that stage. See Candidate A.
9. •<sup>3</sup>, •<sup>4</sup> and •<sup>5</sup> are not available for any attempt to solve a quadratic equation written in the form  $ax^2 + bx = c$  or for erroneously taking out a ‘common factor’ of  $\cos x^\circ$ . See Candidate B.
10. •<sup>4</sup> and •<sup>5</sup> are only available as a consequence of trying to solve a quadratic equation. See Candidate C. However, •<sup>5</sup> is not available if the quadratic has repeated roots or leads to no solutions for  $x$ .
11. Accept  $x = \cos^{-1}\left(\frac{1}{3}\right)$  AND  $x = \cos^{-1}(-2)$  for •<sup>4</sup>.
12. •<sup>5</sup> is only available if one of the equations has no solutions. See Candidate E.
13. Ignore solutions outwith the range.
14. Accept solutions which round to 71 and 289, and  ~~$\cos x^\circ = -2$~~  at •<sup>5</sup>. Do not accept “error” at •<sup>5</sup>.

| Question                                                                                                                                                                                                                                                                                                                                                                    | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                         | Illustrative scheme | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|
| 10. (continued)                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |          |
| <b>Candidate A</b><br>$3(2\cos^2 x^\circ - 1) + 10\cos x^\circ = 1$ • <sup>1</sup> ✓<br>$6\cos^2 x + 10\cos x - 4 = 0$ • <sup>2</sup> ✓<br>$6c^2 + 10c - 4 = 0$<br>$2(3c - 1)(c + 2) = 0$ • <sup>3</sup> ✓<br>$c = \frac{1}{3}, c = -2$ • <sup>4</sup> ✗<br>$x = 70.5, 289.5$ • <sup>5</sup> ✗<br>Where candidates show <del><math>c = -2</math></del> award • <sup>5</sup> | <b>Candidate B - not in standard quadratic equation form</b><br>$3(2\cos^2 x^\circ - 1) + 10\cos x^\circ = 1$ • <sup>1</sup> ✓<br>$6\cos^2 x^\circ + 10\cos x^\circ = 4$ • <sup>2</sup> ✗<br>$2\cos x^\circ(3\cos x^\circ + 5) = 4$ • <sup>3</sup> ✗ • <sup>4</sup> ✗ • <sup>5</sup> ✗<br>$\cos x^\circ = 2, \cos x^\circ = -\frac{1}{3}$<br>No solutions, $x = 109.5, 250.5$                                                          |                     |          |
| <b>Candidate C - not solving a quadratic equation</b><br>$3(2\cos^2 x^\circ - 1) + 10\cos x^\circ = 1$ • <sup>1</sup> ✓<br>$6\cos^2 x^\circ + 10\cos x^\circ - 4 = 0$ • <sup>2</sup> ✓<br>$16\cos x^\circ - 4 = 0$ • <sup>3</sup> ✗ • <sup>4</sup> ✗ • <sup>5</sup> ✗<br>$\cos x^\circ = \frac{1}{4}$<br>$x = 75.5, 284.5$                                                  | <b>Candidate D - reading <math>\cos 2x^\circ</math> as <math>\cos^2 x^\circ</math></b><br>$3\cos^2 x^\circ + 10\cos x^\circ = 1$ • <sup>1</sup> ✗ • <sup>2</sup> ✗<br>$3\cos^2 x^\circ + 10\cos x^\circ - 1 = 0$<br>$\cos x^\circ = \frac{-10 \pm \sqrt{112}}{6}$ • <sup>3</sup> ✓ <sub>1</sub><br>$\cos x^\circ = 0.097, -3.431$ • <sup>4</sup> ✓ <sub>1</sub><br>$x = 84.4, 275.6$ , no solutions      • <sup>5</sup> ✓ <sub>1</sub> |                     |          |
| <b>Candidate E - incorrect factorising</b><br>$\vdots$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$2(\cos x^\circ - 1)(\cos x^\circ + 2) = 0$ • <sup>3</sup> ✗<br>$\cos x^\circ = 1, \cos x^\circ = -2$ • <sup>4</sup> ✓ <sub>1</sub><br>$x = 0$ , no solutions      • <sup>5</sup> ✓ <sub>1</sub>                                                                                | <b>Candidate F - poor notation</b><br>$3(2\cos^2 x^\circ - 1) + 10\cos x^\circ = 1$ • <sup>1</sup> ✓<br>$6\cos^2 + 10\cos x - 4 = 0$ • <sup>2</sup> ^<br>$2(3\cos x^\circ - 1)(\cos x^\circ + 2) = 0$ • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                                    |                     |          |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Max mark |
|----------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.      |  |  | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> state linear equation</li> <li>•<sup>2</sup> introduce logs</li> <li>•<sup>3</sup> use laws of logs</li> <li>•<sup>4</sup> use laws of logs</li> <li>•<sup>5</sup> state <math>a</math> and <math>b</math></li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> state linear equation</li> <li>•<sup>2</sup> convert to exponential form</li> <li>•<sup>3</sup> use laws of indices</li> <li>•<sup>4</sup> state <math>a</math></li> <li>•<sup>5</sup> state <math>b</math></li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_4 y = 2x + 3</math></li> <li>•<sup>2</sup> <math>\log_4 y = 2x \log_4 4 + 3 \log_4 4</math></li> <li>•<sup>3</sup> <math>\log_4 y = \log_4 4^{2x} + \log_4 4^3</math></li> <li>•<sup>4</sup> <math>\log_4 y = \log_4 (4^{2x} \cdot 4^3)</math></li> <li>•<sup>5</sup> <math>a = 64</math> AND <math>b = 16</math></li> </ul> <p>OR</p> $y = 64 \times 16^x$ <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_4 y = 2x + 3</math></li> <li>•<sup>2</sup> <math>y = 4^{2x+3}</math></li> <li>•<sup>3</sup> <math>y = 4^{2x} \cdot 4^3</math></li> <li>•<sup>4</sup> <math>a = 64</math></li> <li>•<sup>5</sup> <math>b = 16</math> OR <math>y = 64 \times 16^x</math></li> </ul> | 5        |
|          |  |  | <p><b>Method 3</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> introduce logs to <math>y = ab^x</math></li> <li>•<sup>2</sup> use laws of logs</li> <li>•<sup>3</sup> interpret intercept</li> <li>•<sup>4</sup> interpret gradient</li> <li>•<sup>5</sup> state <math>a</math> and <math>b</math></li> </ul>                                                                                                                                                                                                                                                                           | <p><b>Method 3</b></p> <p>The equations at •<sup>1</sup>, •<sup>2</sup>, •<sup>3</sup> and •<sup>4</sup> must be stated explicitly</p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_4 y = \log_4 (ab^x)</math></li> <li>•<sup>2</sup> <math>\log_4 y = \log_4 a + x \log_4 b</math></li> <li>•<sup>3</sup> <math>\log_4 a = 3</math></li> <li>•<sup>4</sup> <math>\log_4 b = 2</math></li> <li>•<sup>5</sup> <math>a = 64</math> AND <math>b = 16</math></li> </ul> <p>OR</p> $y = 64 \times 16^x$                                                                                                                                                                                                                                                                                       |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Generic scheme | Illustrative scheme                                                                                                                                                                                                                                       | Max mark |
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| 11. (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |                                                                                                                                                                                                                                                           |          |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |                                                                                                                                                                                                                                                           |          |
| <p>1. In any method, marks may only be awarded within a valid strategy using <math>y = ab^x</math>.</p> <p>2. Accept <math>y = 64.16^x</math> for ●<sup>5</sup> in Methods 1 and 3, and for ●<sup>4</sup> and ●<sup>5</sup> in Method 2.</p> <p>3. Markers must identify the method which best matches the candidate's approach; they must not mix and match between methods.</p> <p>4. Penalise the omission of base 4 at most once in any method.</p> <p>5. Where a candidate uses an incorrect base then only ●<sup>2</sup> and ●<sup>3</sup> are available (in any method).</p> <p>6. Do not penalise the omission of brackets at ●<sup>4</sup> in Method 1 or ●<sup>1</sup> in Method 3.</p> <p>7. For a correct answer with no working, award 1/5.</p> |                |                                                                                                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |                                                                                                                                                                                                                                                           |          |
| <b>Candidate A - Use of coordinate pairs</b><br>$x = 3$ and $\log_4 y = 9$ ● <sup>1</sup> ✓<br>$\Rightarrow x = 3$ and $y = 4^9$ ● <sup>2</sup> ✓<br><br>$x = 0$ and $\log_4 y = 3$<br>$\Rightarrow x = 0$ and $y = 4^3$ ● <sup>3</sup> ✓<br><br>$4^3 = ab^0$ so $a = 64$ ● <sup>4</sup> ✓<br>$4^9 = 64b^3$ so $b = 16$ ● <sup>5</sup> ✓                                                                                                                                                                                                                                                                                                                                                                                                                     |                | <b>Candidate B - ambiguous notation</b><br>$\log_4 y = \log_4 ab^x$ ● <sup>1</sup> ✓<br>$\log_4 y = \log_4 a + \log_4 bx$<br>$\log_4 a = 3$ ● <sup>3</sup> ✓<br>$\log_4 b = 2$ ● <sup>2</sup> ✓ ● <sup>4</sup> ✓<br>$y = 64 \times 16^x$ ● <sup>5</sup> ✓ |          |

| Question                                                                                                                                                                                                                                                                 |     |  | Generic scheme                                                                                                                                                                                                          | Illustrative scheme                                                                                                                                                                                 | Max mark |
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| 12.                                                                                                                                                                                                                                                                      | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> differentiate two non-constant terms</li> <li>•<sup>2</sup> complete differentiation and equate to zero</li> <li>•<sup>3</sup> solve for <math>x</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> eg <math>3x^2 + 4x</math></li> <li>•<sup>2</sup> <math>3x^2 + 4x - 4 = 0</math></li> <li>•<sup>3</sup> <math>\frac{2}{3}, -2</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                            |     |  |                                                                                                                                                                                                                         |                                                                                                                                                                                                     |          |
| 1. For a numerical approach, award 0/3.<br>2. • <sup>2</sup> is only available if '= 0' appears at • <sup>2</sup> stage or in the working leading to • <sup>3</sup> . However, see Candidate A.<br>3. • <sup>3</sup> is only available for solving a quadratic equation. |     |  |                                                                                                                                                                                                                         |                                                                                                                                                                                                     |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                      |     |  |                                                                                                                                                                                                                         |                                                                                                                                                                                                     |          |
| <b>Candidate A - derivative not explicitly equated to 0</b><br>Stationary points occur when $f'(x) = 0$<br>$f'(x) = 3x^2 + 4x - 4$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$f'(x) = (3x - 2)(x + 2)$<br>$x = \frac{2}{3}, -2$ • <sup>3</sup> ✓                              |     |  | <b>Candidate B - derivative never equated to 0</b><br>$3x^2 + 4x - 4$ • <sup>1</sup> ✓ • <sup>2</sup> ^<br>$(3x - 2)(x + 2)$<br>$x = \frac{2}{3}, -2$ • <sup>3</sup> ✓                                                  |                                                                                                                                                                                                     |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | Generic scheme                                                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                                                                                                                                                                                             | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> evaluate <math>f(x)</math> at stationary point</li> <li>•<sup>5</sup> consider value of <math>f(x)</math> at end points</li> <li>•<sup>6</sup> state maximum <b>AND</b> minimum values</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>4</sup> <math>-\frac{13}{27}</math> or <math>-0.481\dots</math> or <math>\left(\frac{2}{3}, -\frac{13}{27}\right)</math></li> <li>•<sup>5</sup> <math>f(1) = 0</math>, <math>f(-1) = 6</math></li> <li>•<sup>6</sup> max (value) = 6</li> </ul> <p><b>AND</b></p> <p>min (value) = <math>-\frac{13}{27}</math></p> | 3        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |  |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                 |          |
| <p>4. 'Greatest <math>(-1, 6)</math>; least <math>\left(\frac{2}{3}, -\frac{13}{27}\right)</math>', does not gain •<sup>6</sup>.</p> <p>5. •<sup>4</sup> is not available for only evaluating <math>y</math> at a value of <math>x</math>, obtained at the •<sup>3</sup> stage, which lies outwith the interval <math>-1 \leq x \leq 1</math>.</p> <p>6. •<sup>5</sup> is not available for evaluating <math>y</math> at an end point which is also identified as a stationary point in part (a).</p> <p>7. •<sup>6</sup> is only available where candidates have attempted to evaluate <math>y</math> at stationary point(s) in the interval <math>-1 \leq x \leq 1</math> identified in part (a).</p> <p>8. •<sup>6</sup> is only available where candidates have attempted to evaluate <math>y</math> at <math>x = -1</math> and <math>x = 1</math>.</p> |     |  |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                 |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |  |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                 |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                 |          |

| Question |     |  | Generic scheme                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                                                                                                                                                                        | Max mark |
|----------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 13.      | (a) |  | <p>•<sup>1</sup> use compound angle formula</p> <p>•<sup>2</sup> compare coefficients</p> <p>•<sup>3</sup> process for <math>k</math></p> <p>•<sup>4</sup> process for <math>a</math> and express in required form</p> | <p>•<sup>1</sup> <math>k \sin x^\circ \cos a^\circ - k \cos x^\circ \sin a^\circ</math> stated explicitly</p> <p>•<sup>2</sup> <math>k \cos a^\circ = 2</math> AND <math>k \sin a^\circ = 7</math> stated explicitly</p> <p>•<sup>3</sup> <math>k = \sqrt{53}</math></p> <p>•<sup>4</sup> <math>\sqrt{53} \sin(x - 74.05)^\circ</math></p> | 4        |

#### Notes:

- Do not penalise the omission of degree symbols.
- Accept  $k(\sin x^\circ \cos a^\circ - \cos x^\circ \sin a^\circ)$  for •<sup>1</sup>.
- Treat  $k \sin x^\circ \cos a^\circ - \cos x^\circ \sin a^\circ$  as bad form only if the equations at the •<sup>2</sup> stage both contain  $k$ .
- $\sqrt{53} \sin x^\circ \cos a^\circ - \sqrt{53} \cos x^\circ \sin a^\circ$  or  $\sqrt{53}(\sin x^\circ \cos a^\circ - \cos x^\circ \sin a^\circ)$  are acceptable for •<sup>1</sup> and •<sup>3</sup>.
- <sup>2</sup> is not available for  $k \sin x^\circ = 7$  and  $k \cos x^\circ = 2$ . However, •<sup>4</sup> may still be gained. See Candidate F.
- Accept  $-k \sin a^\circ = -7$  and  $k \cos a^\circ = 2$  for •<sup>2</sup>.
- <sup>4</sup> is not available for a value of  $a$  given in radians.
- Accept values of  $a$  which round to 74.
- Candidates may **state and use** any form of the wave function for •<sup>1</sup>, •<sup>2</sup> and •<sup>3</sup>.  
However, •<sup>4</sup> is only available if the wave is interpreted in the form  $k \sin(x - a)^\circ$ .
- Evidence for •<sup>4</sup> may not appear until part (b) and must appear by the •<sup>5</sup> stage.

#### Commonly Observed Responses:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <p><b>Candidate A</b></p> <p><math>\sqrt{53} \cos a^\circ = 2</math>      •<sup>1</sup> ^</p> <p><math>\sqrt{53} \sin a^\circ = 7</math>      •<sup>2</sup> ✓ •<sup>3</sup> ✓</p> <p><math>\tan a^\circ = \frac{7}{2}</math></p> <p><math>a = 74.05\dots</math></p> <p><math>\sqrt{53} \sin(x - 74.05\dots)^\circ</math>      •<sup>4</sup> ✓</p>                                                                                                                                                                        | <p><b>Candidate B</b></p> <p><math>k \sin x^\circ \cos a^\circ - k \cos x^\circ \sin a^\circ</math>      •<sup>1</sup> ✓</p> <p><math>\cos a^\circ = 2</math></p> <p><math>\sin a^\circ = 7</math>      •<sup>2</sup> ✗</p> <p><math>\tan a^\circ = \frac{7}{2}</math></p> <p><math>a = 74.05\dots</math> (not consistent with equations at •<sup>2</sup>)</p> <p><math>\sqrt{53} \sin(x - 74.05\dots)^\circ</math>      •<sup>3</sup> ✓ •<sup>4</sup> ✗</p> |
| <p><b>Candidate C - missing <math>k</math></b></p> <p><math>\sin x^\circ \cos a^\circ - \cos x^\circ \sin a^\circ</math>      •<sup>1</sup> ✗</p> <p><math>\cos a^\circ = 2</math></p> <p><math>\sin a^\circ = 7</math>      •<sup>2</sup> ✗</p> <p><math>k = \sqrt{53}</math>      •<sup>3</sup> ✓</p> <p><math>\tan a^\circ = \frac{7}{2}</math></p> <p><math>a = 74.05\dots</math> (not consistent with equations at •<sup>2</sup>)</p> <p><math>\sqrt{53} \sin(x - 74.05\dots)^\circ</math>      •<sup>4</sup> ✗</p> | <p><b>Candidate D - errors at •<sup>2</sup></b></p> <p><math>k \sin x^\circ \cos a^\circ - k \cos x^\circ \sin a^\circ</math>      •<sup>1</sup> ✓</p> <p><math>k \cos a^\circ = 7</math></p> <p><math>k \sin a^\circ = 2</math>      •<sup>2</sup> ✗</p> <p><math>\tan a^\circ = \frac{2}{7}</math></p> <p><math>a = 15.94\dots</math></p> <p><math>\sqrt{53} \sin(x - 15.94\dots)^\circ</math>      •<sup>3</sup> ✓ •<sup>4</sup> ✓<sub>1</sub></p>        |

| Question                                                                                                                                                                                                                                                      | Generic scheme                                                                                 | Illustrative scheme                                                                                                                                                                                                                                                      | Max mark                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>13. (a) (continued)</b>                                                                                                                                                                                                                                    |                                                                                                |                                                                                                                                                                                                                                                                          |                                                                                                |
| <b>Commonly Observed Responses (continued):</b>                                                                                                                                                                                                               |                                                                                                |                                                                                                                                                                                                                                                                          |                                                                                                |
| <b>Candidate E - errors at ●<sup>2</sup></b><br>$k \sin x^\circ \cos a^\circ - k \cos x^\circ \sin a^\circ$<br>$k \cos a^\circ = 2$<br>$k \sin a^\circ = -7$<br>$\tan a^\circ = -\frac{7}{2}$<br>$a = 285.94\dots$<br>$\sqrt{53} \sin(x - 285.94\dots)^\circ$ | ● <sup>1</sup> ✓<br><br>● <sup>2</sup> ✗<br><br>● <sup>3</sup> ✓ ● <sup>4</sup> ✓ <sub>1</sub> | <b>Candidate F - use of <math>x</math> at ●<sup>2</sup></b><br>$k \sin x^\circ \cos a^\circ - k \cos x^\circ \sin a^\circ$<br>$k \cos x^\circ = 2$<br>$k \sin x^\circ = 7$<br>$\tan a^\circ = \frac{7}{2}$<br>$a = 74.05\dots$<br>$\sqrt{53} \sin(x - 74.05\dots)^\circ$ | ● <sup>1</sup> ✓<br><br>● <sup>2</sup> ✗<br><br>● <sup>3</sup> ✓ ● <sup>4</sup> ✓ <sub>1</sub> |
| <b>Candidate G</b><br>$k \sin A \cos B - k \cos B \sin A$<br>$k \cos A = 2$<br>$k \sin A = 7$<br>$\tan A = \frac{7}{2}$<br>$A = 74.05\dots$<br>$\sqrt{53} \sin(x - 74.05\dots)^\circ$                                                                         | ● <sup>1</sup> ✗<br><br>● <sup>2</sup> ✗<br><br>● <sup>3</sup> ✓ ● <sup>4</sup> ✓ <sub>1</sub> | <b>Candidate H - missing <math>k</math></b><br>$\sin x^\circ \cos a^\circ - \cos x^\circ \sin a^\circ$<br>$k \cos a^\circ = 2$<br>$k \sin a^\circ = 7$<br>$\tan a^\circ = \frac{7}{2}$<br>$a = 74.05\dots$<br>$\sqrt{53} \sin(x - 74.05\dots)^\circ$                     | ● <sup>1</sup> ✗<br><br>● <sup>2</sup> ✗<br><br>● <sup>3</sup> ✓ ● <sup>4</sup> ✓ <sub>1</sub> |



| Question                                                                                                                                                                                                                                                                 |     |  | Generic scheme                                                                                                                                                                                                                    | Illustrative scheme                                                                                                                                                                                                                                                           | Max mark |
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| 13.                                                                                                                                                                                                                                                                      | (b) |  | <ul style="list-style-type: none"> <li>•<sup>5</sup> form equation, link to (a) and rearrange</li> <li>•<sup>6</sup> solve for <math>x - 74.05 \dots</math></li> <li>•<sup>7</sup> find <math>x</math>-coordinate of P</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>5</sup> <math>\sin(x - 74.05)^\circ = \frac{3}{\sqrt{53}}</math></li> <li>•<sup>6</sup> <math>x - 74.05 = 24.33</math> OR <math>155.66</math><br/>OR <math>-204.33</math></li> <li>•<sup>7</sup> <math>-130.28</math></li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                            |     |  |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                               |          |
| 11. Accept any value of $x$ which rounds to $-130$ .<br>12. • <sup>7</sup> is only available for a single value of $x$ . See Candidate L.<br>13. When awarding follow-through marks, • <sup>7</sup> is only available for a single value of $x$ between $-180$ and $0$ . |     |  |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                               |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                      |     |  |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                               |          |
| <b>Candidate I - Follow through from COR D</b>                                                                                                                                                                                                                           |     |  | <b>Candidate J - Follow through from COR E</b>                                                                                                                                                                                    |                                                                                                                                                                                                                                                                               |          |
| $\sin(x - 15.94\dots)^\circ = \frac{3}{\sqrt{53}}$                                                                                                                                                                                                                       |     |  | $\sin(x - 285.94\dots)^\circ = \frac{3}{\sqrt{53}}$                                                                                                                                                                               |                                                                                                                                                                                                                                                                               |          |
| $x - 15.94 = 24.33$ (or $155.66$ )                                                                                                                                                                                                                                       |     |  | $x - 285.94 = 24.33$ (or $155.66$ )                                                                                                                                                                                               |                                                                                                                                                                                                                                                                               |          |
| $x = -188.39\dots$                                                                                                                                                                                                                                                       |     |  | $x = -49.71\dots$                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                               |          |
| <b>Candidate K - Missing rearrangement</b>                                                                                                                                                                                                                               |     |  | <b>Candidate L - multiple solutions</b>                                                                                                                                                                                           |                                                                                                                                                                                                                                                                               |          |
| $\sqrt{53} \sin(x - 74.05)^\circ = 3$                                                                                                                                                                                                                                    |     |  | $\sin(x - 74)^\circ = \frac{3}{\sqrt{53}}$                                                                                                                                                                                        |                                                                                                                                                                                                                                                                               |          |
| $x - 74.05 = 24.33$                                                                                                                                                                                                                                                      |     |  | $x - 74.05 = 24.33, 155.66, -204.33$                                                                                                                                                                                              |                                                                                                                                                                                                                                                                               |          |
|                                                                                                                                                                                                                                                                          |     |  | $x = 98.39, 229.72, -130.28$                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                               |          |
|                                                                                                                                                                                                                                                                          |     |  | However, where the final solution is clearly identified by the candidate, award • <sup>7</sup>                                                                                                                                    |                                                                                                                                                                                                                                                                               |          |

| Question                     |     |  | Generic scheme                       | Illustrative scheme     | Max mark |
|------------------------------|-----|--|--------------------------------------|-------------------------|----------|
| 14.                          | (a) |  | • <sup>1</sup> calculate temperature | • <sup>1</sup> 825 (°C) | 1        |
| Notes:                       |     |  |                                      |                         |          |
|                              |     |  |                                      |                         |          |
| Commonly Observed Responses: |     |  |                                      |                         |          |
|                              |     |  |                                      |                         |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  | Generic scheme                                                                                                                                                                                                                           | Illustrative scheme                                                                                                                                                                                                                                                                                     | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 14.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (b) |  | <ul style="list-style-type: none"> <li>•<sup>2</sup> substitute for <math>T</math></li> <li>•<sup>3</sup> process equation</li> <li>•<sup>4</sup> express in logarithmic form</li> <li>•<sup>5</sup> solve for <math>t</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>2</sup> <math>150 = 800 \times e^{-0.124t} + 25</math></li> <li>•<sup>3</sup> <math>\frac{150 - 25}{800} = e^{-0.124t}</math></li> <li>•<sup>4</sup> <math>\log_e \frac{150 - 25}{800} = -0.124t</math></li> <li>•<sup>5</sup> 14.970 (seconds)</li> </ul> | 4        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                         |          |
| <p>1. Where values other than 150 are used in the substitution •<sup>3</sup>, •<sup>4</sup> and •<sup>5</sup> are still available.</p> <p>2. •<sup>3</sup> may be implied by •<sup>4</sup>.</p> <p>3. Evidence for •<sup>4</sup> must be stated explicitly. See Candidate B.</p> <p>4. At •<sup>4</sup> all exponentials must be processed.</p> <p>5. Any base may be used at •<sup>4</sup> stage, See Candidate A.</p> <p>6. Accept <math>\ln 0.15625 = -0.124t \ln e</math> and <math>-1.856... = -0.124t</math> for •<sup>4</sup>.</p> <p>7. •<sup>5</sup> is unavailable where candidates round the value of <math>\ln 0.15625</math> to fewer than 2 decimal places.</p> <p>8. Accept answers where <math>14.9 \leq t \leq 15</math> at •<sup>5</sup>.</p> <p>9. The calculation at •<sup>5</sup> must follow from the valid use of exponentials and logarithms at •<sup>3</sup> and •<sup>4</sup>.</p> <p>10. Where candidates show no working, or take an iterative approach to arrive at <math>t = 14</math> or <math>t = 15</math>, award 1/4. However, if, in any iterations <math>T</math> is evaluated for <math>t = 14.9</math> and <math>t = 15</math>, leading to a final answer of <math>t = 15</math> (seconds), then award 4/4. See Candidate C.</p> |     |  |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                         |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |  |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                         |          |
| Candidate A - using other bases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |  | Candidate B - missing working                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                         |          |
| $150 = 800 \times e^{-0.124t} + 25$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | $150 = 800 \times e^{-0.124t} + 25$                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                         |          |
| $\frac{5}{32} = e^{-0.124t}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |  | $0.15625 = e^{-0.124t}$                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                         |          |
| $\log_{10} \frac{5}{32} = -0.124t \log_{10} e$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |  | $t = 14.97$                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                         |          |
| $t = 14.97$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                         |          |
| Candidate C - iterative approach                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |  | Candidate D - taking logarithms of both sides                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                         |          |
| $t = 13 \Rightarrow T = 184.59...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  | $150 = 800 \times e^{-0.124t} + 25$                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                         |          |
| $t = 14 \Rightarrow T = 165.979...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | $125 = 800 \times e^{-0.124t}$                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                         |          |
| $t = 14.9 \Rightarrow T = 151.091...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |  | $\log_e 125 = \log_e (800 \times e^{-0.124t})$                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                         |          |
| $t = 15 \Rightarrow T = 149.538...$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |  | $\log_e 125 = \log_e 800 + \log_e e^{-0.124t}$                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                         |          |
| so $t = 15$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | $\log_e 125 - \log_e 800 = -0.124t$                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                         |          |
| Award 4/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |  | $t = 14.97$                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                         |          |
| Candidate E - transcription error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |  | Candidate F - Incorrect processing                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                         |          |
| $150 = 800e^{-0.124t}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  | $150 = 800 \times e^{-0.124t} + 25$                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                         |          |
| $0.1875 = e^{-0.124t}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  | $\log_e 150 = \log_e 800 \times e^{-0.124t} + 25$                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                         |          |
| $-0.124t = -1.673$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |  | $\frac{\log_e 150 - 25}{\log_e 800} = e^{-0.124t}$                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                         |          |
| $t = 13.5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                         |          |

| Question                                                                                                                                                                                                                                                                                       | Generic scheme | Illustrative scheme | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|----------|
| 14. (b) (continued)                                                                                                                                                                                                                                                                            |                |                     |          |
| Commonly Observed Responses (continued):                                                                                                                                                                                                                                                       |                |                     |          |
| <b>Candidate G - Incorrect processing</b><br>$150 = 800 \times e^{-0.124t} + 25$ • <sup>2</sup> ✓<br>$\frac{150}{800} = e^{-0.124t} + 25$ • <sup>3</sup> ✗ • <sup>4</sup> ✗ • <sup>5</sup> ✗<br>$\frac{150}{800} - 25 = e^{-0.124t}$<br>$\log_e \left( \frac{150}{800} - 25 \right) = -0.124t$ |                |                     |          |

| Question                                                                                                                                                                                                                                                                                                                                                   |  |  | Generic scheme                                                                                                                                                                                                             | Illustrative scheme                                                                                                                                                                                                                                                                            | Max mark                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 15.                                                                                                                                                                                                                                                                                                                                                        |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> expand</li> <li>•<sup>2</sup> use appropriate trigonometric identity</li> <li>•<sup>3</sup> use second trigonometric identity and express in required form</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>3\sin^2\theta + 9\sin\theta\cos\theta + \cos\theta\sin\theta + 3\cos^2\theta</math></li> <li>•<sup>2</sup> <math>3\dots</math> or <math>\dots 5\sin 2\theta</math></li> <li>•<sup>3</sup> <math>3 + 5\sin 2\theta</math></li> </ul> | 3                             |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                              |  |  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                |                               |
| <p>1. •<sup>1</sup> and •<sup>2</sup> are available to candidates work with, for example, A or <math>x</math> in place of <math>\theta</math>. However, •<sup>3</sup> is not available.</p> <p>2. Marks are not available to candidates for quoting <math>\sin^2\theta + \cos^2\theta = 1</math> or <math>\sin 2\theta = 2\sin\theta\cos\theta</math>.</p> |  |  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                |                               |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                        |  |  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                |                               |
| <b>Candidate A - incorrect notation</b>                                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                                                                            | <b>Candidate B - incorrect notation</b>                                                                                                                                                                                                                                                        |                               |
| $3\sin^2\theta + 10\sin\theta\cos\theta + 3\cos^2\theta$                                                                                                                                                                                                                                                                                                   |  |  | • <sup>1</sup> ✗                                                                                                                                                                                                           | $3\sin^2\theta + 10\sin\theta\cos\theta + 3\cos^2\theta$                                                                                                                                                                                                                                       | • <sup>1</sup> ✗              |
| $3 + 10\sin\theta\cos\theta$                                                                                                                                                                                                                                                                                                                               |  |  | • <sup>2</sup> ✗                                                                                                                                                                                                           | $3 + 10\sin\theta\cos\theta$                                                                                                                                                                                                                                                                   | • <sup>2</sup> ✗              |
| $3 + 5\sin 2\theta$                                                                                                                                                                                                                                                                                                                                        |  |  | • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                              | $3 + 5\sin 2\theta$                                                                                                                                                                                                                                                                            | • <sup>3</sup> ✓ <sub>1</sub> |
| <b>Candidate C - incorrect expansion leading to a non-integer value of <math>b</math></b>                                                                                                                                                                                                                                                                  |  |  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                |                               |
| $3\sin^2\theta + 6\sin\theta\cos\theta + \cos\theta\sin\theta + 3\cos^2\theta$                                                                                                                                                                                                                                                                             |  |  | • <sup>1</sup> ✗                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                |                               |
| $3\sin^2\theta + 7\sin\theta\cos\theta + 3\cos^2\theta$                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                |                               |
| $3 + 7\sin\theta\cos\theta$                                                                                                                                                                                                                                                                                                                                |  |  | • <sup>2</sup> ✓ <sub>1</sub>                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                |                               |
| $3 + 3.5\sin 2\theta$                                                                                                                                                                                                                                                                                                                                      |  |  | • <sup>3</sup> ✗                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                |                               |

[END OF MARKING INSTRUCTIONS]

## General marking principles for Higher Mathematics

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

*For each question, the marking instructions are generally in two sections:*

- *generic scheme – this indicates why each mark is awarded*
- *illustrative scheme – this covers methods which are commonly seen throughout the marking*

*In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.*

- Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.
- One mark is available for each •. There are no half marks.
- If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.
- Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.
- Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.
- If an error is trivial, casual or insignificant, for example  $6 \times 6 = 12$ , candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.
- If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 1 \text{ or } 3$$

(i) Horizontal/vertical marking

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

$$\begin{array}{cc} \bullet^5 & \bullet^6 \\ \bullet^5 & x = 2 \quad x = -4 \\ \bullet^6 & y = 5 \quad y = -7 \end{array}$$

Horizontal:

$$\begin{array}{l} \bullet^5 \quad x = 2 \text{ and } x = -4 \\ \bullet^6 \quad y = 5 \text{ and } y = -7 \end{array}$$

Vertical:

$$\begin{array}{l} \bullet^5 \quad x = 2 \text{ and } y = 5 \\ \bullet^6 \quad x = -4 \text{ and } y = -7 \end{array}$$

You must choose whichever method benefits the candidate, not a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$$\frac{15}{12} \text{ must be simplified to } \frac{5}{4} \text{ or } 1\frac{1}{4} \quad \frac{43}{1} \text{ must be simplified to } 43$$

$$\frac{15}{0.3} \text{ must be simplified to } 50 \quad \frac{4\cancel{5}}{3} \text{ must be simplified to } \frac{4}{15}$$

$$\sqrt{64} \text{ must be simplified to } 8^*$$

\*The square root of perfect squares up to and including 144 must be known.

(k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

(l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example

$$(x^3 + 2x^2 + 3x + 2)(2x + 1) \text{ written as}$$

$$(x^3 + 2x^2 + 3x + 2) \times 2x + 1$$

$$= 2x^4 + 5x^3 + 8x^2 + 7x + 2$$

gains full credit

- repeated error within a question, but not between questions or papers

(m) In any ‘Show that...’ question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

- (n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.
- (o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.
- (p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks.                             | Strategy 2 attempt 1 is worth 1 mark.                              |
| Strategy 1 attempt 2 is worth 4 marks.                             | Strategy 2 attempt 2 is worth 5 marks.                             |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.